Free Standing Enclosure, phase 1 (no tanks, no backlight): DRAFT modular order

Scene `free-standing-enclosure-991056`: Free Standing Enclosure (Vault file-version 991056) native snapshot. Document `free-standing-enclosure-phase-1-draft`. Generated 2026-10-23T15:40:18Z.

DRAFT

Review state: draft. This sequence has not been physically validated and the order is a draft to be corrected after the build.

Build scope phase-1: Enclosure without tanks and backlight (decided by Peter, 2026-10-22; draft). 110 of 155 installed occurrences are in this build, 42 conditional (undecided), 3 deferred; CAD counts are unchanged and the deferred items stay in the model.

Phase note: Phase 1 assembles the frame, casters, panels with handles, toggle clamps, lower supports, camera frame and top panel. Deferred items stay in the CAD scene and in the CAD counts; this file only scopes the build and its documentation.

Backlight: no modeled part is identified as the backlight or light box; nothing further is excluded until Peter names the occurrences

Deferred design: monitor supports and acrylic; bowls and bowl frame; tank frame changes.

Geometry and hierarchy come from the CAD export named in the build record. Quantities are CAD instance counts, not purchase quantities, kits, stock or an approved bill of materials.

Project note: The tank(s) and the lower light box are to be replaced in the modified build by two Alienware AW2725Q monitors under an acrylic panel and bowl frames (Peter, 2026-10-22; intent, not a designer confirmation; the light-box occurrences are not yet identified). CAD counts are unchanged.

Subassembly pages and joint close-ups are inspection views of parts at their CAD positions; they are not assembly steps and imply no sequence, fasteners or tool access.

Supplementary items (20 rows, 16 for the existing design, 4 for new design in the appendix only) are hardware not modelled in CAD: listed separately for review, never added to CAD quantities, unknown quantities left unknown; not an approved bill of materials.

CAD ambiguity: 1 pair(s) of occurrences of the same part share one position (SHULZE 2W.353 TANK). All source instances are kept and counted; review the intended configuration before treating the count as stock.

Approach arrows are unverified render hints; they do not prove an insertion path or tool access.

Stock is unknown until counted and recorded (on_hand_each with counted_by and counted_date). The counted box is for a paper count only.

Finished assembly

Figure 1: All parts at their CAD positions.

Subassemblies, drawn apart

exploded illustration: the top-level groups moved apart as rigid bodies by the recorded offsets; not an assembly step, no order, fastener or tool access implied. The numbers are the subassembly pages of this document, one callout per subassembly on one of its instances.

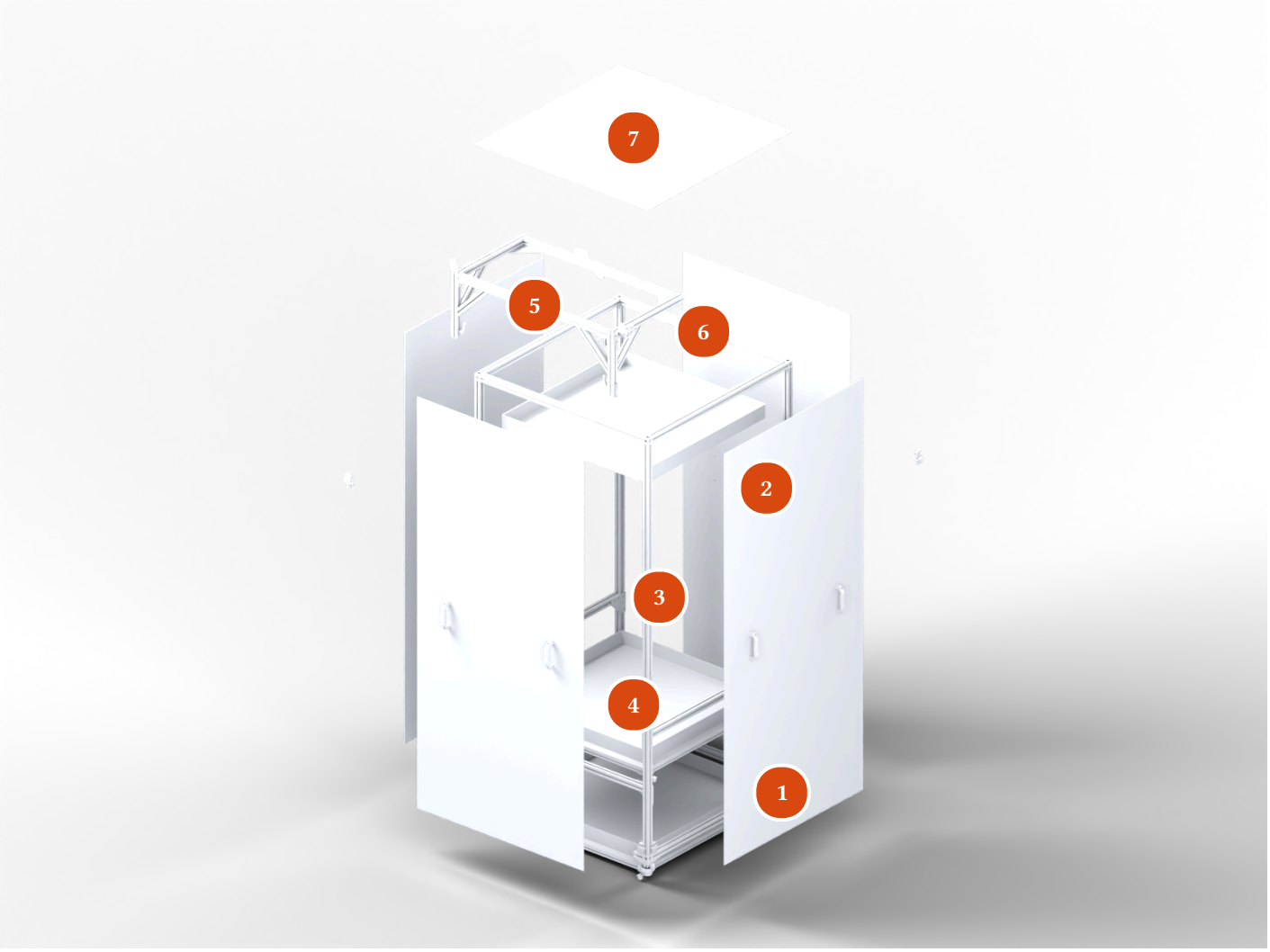


Illustration offsets only (recorded in manual.json); nothing here is an assembly order.

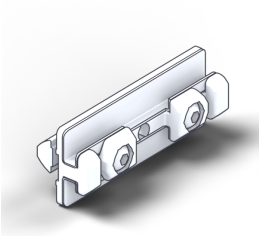
#	Subassembly	Instances	Page
1	**CASTER ASSEMBLY** (4 in this project) callout on `FREE STANDING FRAME:1/CASTER ASSEMBLY:2`	`FREE STANDING FRAME:1/CASTER ASSEMBLY:1`, `FREE STANDING FRAME:1/CASTER ASSEMBLY:2`, `FREE STANDING FRAME:1/CASTER ASSEMBLY:3`, `FREE STANDING FRAME:1/CASTER ASSEMBLY:4`	page 14
2	**PANEL ASSEMBLY** (4 in this project) callout on `PANEL ASSEMBLY:3`	`PANEL ASSEMBLY:1`, `PANEL ASSEMBLY:2`, `PANEL ASSEMBLY:3`, `PANEL ASSEMBLY:4`	page 15
3	**TOGGLE CLAMP ASSEMBLY** (4 in this project) callout on `TOGGLE CLAMP ASSEMBLY:4`	`TOGGLE CLAMP ASSEMBLY:1`, `TOGGLE CLAMP ASSEMBLY:2`, `TOGGLE CLAMP ASSEMBLY:3`, `TOGGLE CLAMP ASSEMBLY:4`	page 16
4	**TANK SUPPORT FRAME** (2 in this project) callout on `TANK SUPPORT FRAME:1`	`TANK SUPPORT FRAME:1`, `TANK SUPPORT FRAME:2`	page 18
5	**CAMERA FRAME** (1 in this project) callout on `CAMERA FRAME:1`	`CAMERA FRAME:1`	page 20
6	**FREE STANDING FRAME** (1 in this project) callout on `FREE STANDING FRAME:1`	`FREE STANDING FRAME:1`	page 22

#	Subassembly	Instances	Page
7	**Parts directly under the root** 34 ENCLOSURE TOP PANEL ×1 37 SHULZE 2W.353 TANK ×1	`SHULZE 2W.353 TANK:1`, `ENCLOSURE TOP PANEL:1`	see Parts

Parts

CAD qty is the number of instances of the part in the CAD model (reference-only geometry excluded). It is not a purchase quantity: purchased kits, pack sizes, spares and stock are not reconciled. The large number is the quantity for phase-1 (this build); conditional and deferred quantities are stated beside it. Write the quantity you count in the box.

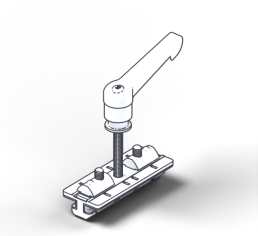
1 MCMASTER 5537T293 T-Slotted Framing



overall 80 × 32 × 28.4 mm
Zinc · 9 bodies
on hand unknown
intended configuration confirmed: no

2 this build 2 — counted ☐
CAD qty 2 each

2 MCMASTER 5537T429 T-Slotted Framing



overall 103 × 99.2 × 32 mm
Zinc · 11 bodies
on hand unknown
intended configuration confirmed: no

2 this build 2 — counted ☐
CAD qty 2 each

3 SLLB Enclosure 15in 47065T808 T-Slotted Framing

overall 381 × 38.1 × 38.1 mm
ALUMINUM, 6061-T651
on hand unknown
intended configuration confirmed: no

2 this build 2 — counted ☐
CAD qty 2 each

4 SLLB Enclosure 22in 47065T808 T-Slotted Framing

overall 559 × 38.1 × 38.1 mm
ALUMINUM, 6061-T651
on hand unknown
intended configuration confirmed: no

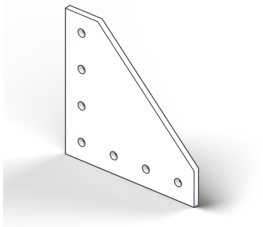
2 this build 2 — counted ☐
CAD qty 2 each

5 SLLB Enclosure 41in 47065T808 T-Slotted Framing

overall 1032 × 38.1 × 38.1 mm
ALUMINUM, 6061-T651
on hand unknown
intended configuration confirmed: no

2 this build 2 — counted ☐
CAD qty 2 each

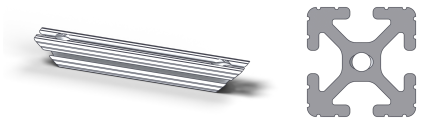
6 MCMASTER 3136N149 T-Slotted Framing



overall 152 × 152 × 6.3 mm
ALUMINUM, 6061-T651
on hand unknown
intended configuration confirmed: no

2 this build 2 — counted ☐
CAD qty 2 each

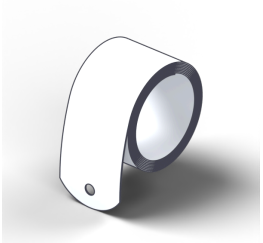
7 MCMASTER 47065T27 T-Slotted Framing



overall 305 × 38.1 × 38.1 mm
ALUMINUM, 6061-T651
on hand unknown
intended configuration confirmed: no

4 this build 4
CAD qty 4 each
counted ☐

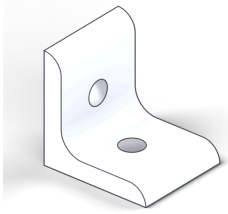
8 MCMASTER 9293K121 Constant-Force Spring



overall 59.9 × 52.6 × 31.8 mm
Generic
on hand unknown
intended configuration confirmed: no

2 this build 2
CAD qty 2 each
counted ☐

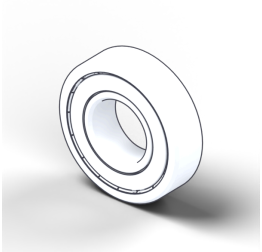
9 MCMASTER 47065T845 Silver Corner Bracket



overall 38.1 × 38.1 × 33.3 mm
ALUMINUM, 6061-T651
on hand unknown
intended configuration confirmed: no

4 this build 4
CAD qty 4 each
counted ☐

10 MCMASTER 5972K164 Ball Bearing



overall 22 × 22 × 6 mm
P/N MCMASTER 5972K164_Ball Bearing · Generic · 14 bodies
on hand unknown
intended configuration confirmed: no

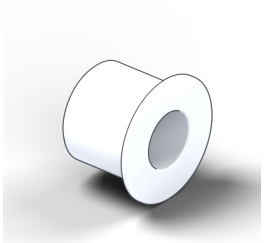
4 this build 4
CAD qty 4 each
counted ☐

11 MCMASTER 90265A164 18-8 Stainless Steel Shoulder Screw

overall 45.3 × 16 × 16 mm
P/N MCMASTER 90265A164_18-8 Stainless Steel Shoulder Screw · Generic
on hand unknown
intended configuration confirmed: no

2 this build 2
CAD qty 2 each
counted ☐

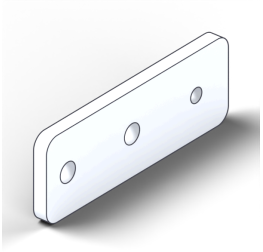
12 SPRING ROLLER



overall 48 × 48 × 34.3 mm (1.89 × 1.89 × 1.35 in)
ABS, BLACK
on hand unknown
intended configuration confirmed: no

2 this build 2
CAD qty 2 each
counted ☐

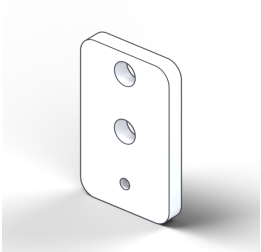
13 ROLLER MOUNT PLATE



overall 108 × 38.1 × 6.3 mm (4.25 × 1.5 × 0.25 in)
ALUMINUM, 6061-T651
on hand unknown
intended configuration confirmed: no

2 this build 2
CAD qty 2 each
counted ☐

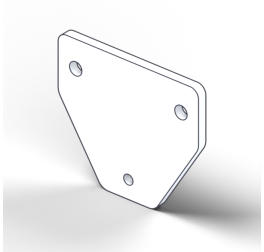
14 SPRING MOUNT PLATE



overall 60 × 38.1 × 6.3 mm (2.362 × 1.5 × 0.25 in)
P/N SPRING MOUNT PLATE · ALUMINUM, 6061-T651
on hand unknown
intended configuration confirmed: no

2 this build 2
CAD qty 2 each
counted ☐

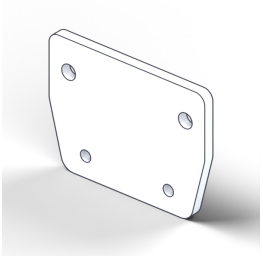
15 CAMERA MOUNT PLATE



overall 102 × 88.9 × 6.3 mm (4 × 3.5 × 0.25 in)
P/N CAMERA MOUNT PLATE · ALUMINUM, 6061-T651
on hand unknown
intended configuration confirmed: no

1 this build 1
CAD qty 1 each
counted ☐

16 CAMERA MOUNT PLATE - TELEDYNE KINETIX



overall 102 × 76.2 × 6.3 mm (4 × 3 × 0.25 in)
ALUMINUM, 6061-T651
on hand unknown
intended configuration confirmed: no

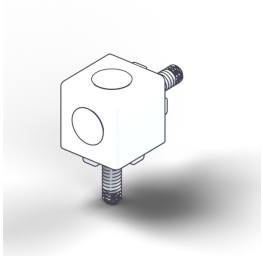
1 this build 1 counted
CAD qty 1 each

17 TSLOT EXTRUSION 1X40.5 copy

overall 1029 × 25.4 × 25.4 mm (40.5 × 1 × 1 in)
Generic
on hand unknown
intended configuration confirmed: no

0 this build 0 counted
CAD qty 8 each · conditional 8

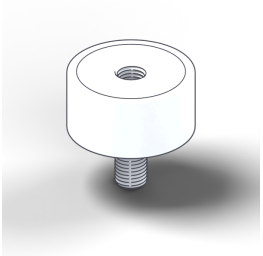
18 T SLOT CORNER 47065T242



overall 47.4 × 47.4 × 25.2 mm (1.868 × 1.868 × 0.992 in)
P/N 47065T242_T-Slotted Framing · Generic · 5 bodies
on hand unknown
intended configuration confirmed: no

0 this build 0 counted
CAD qty 8 each · conditional 8

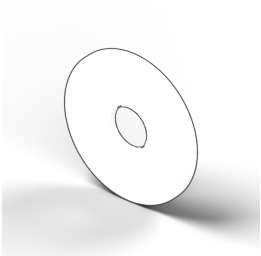
19 MCMASTER 9378K24 Natural Rubber Vibration-Damping Sandwich



overall 25.7 × 25.4 × 25.4 mm (1.013 × 1 × 1 in)
P/N 9378K24_Natural Rubber Vibration-Damping Sandwich Mount · Generic · 3 bodies
on hand unknown
intended configuration confirmed: no

0 this build 0 counted
CAD qty 8 each · conditional 8

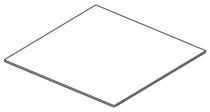
20 MCMASTER 90313A107 18-8 Stainless Steel Oversized Washer



overall 25.4 × 25.4 × 1.3 mm (1 × 1 × 0.05 in)
P/N 90313A107_18-8 Stainless Steel Oversized Washer · STAINLESS STEEL, 18-8
on hand unknown
intended configuration confirmed: no

0 this build 0 counted
CAD qty 8 each · conditional 8

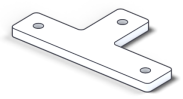
21 TANK SUPPORT PLATFORM



overall 1080 × 1080 × 12.7 mm (42.5 × 42.5 × 0.5 in)
P/N TANK SUPPORT PLATFORM · ACRYLIC, CLEAR
on hand unknown
intended configuration confirmed: no

0 this build 0 counted
CAD qty 2 each · conditional 2

22 TANK SUPPORT BRACKET



overall 114 × 63.5 × 6.3 mm (4.5 × 2.5 × 0.25 in)
ALUMINUM, 6061-T651
on hand unknown
intended configuration confirmed: no

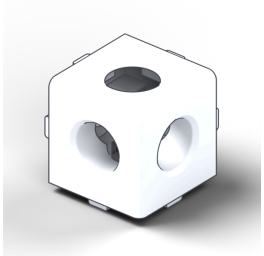
0 this build 0 counted
CAD qty 8 each · conditional 8

23 SLLB Enclosure 44in 47065T808 T-Slotted Framing

overall 1118 × 38.1 × 38.1 mm
P/N SLLB Enclosure 44in 47065T808_T-Slotted Framing · Generic
on hand unknown
intended configuration confirmed: no

11 this build 11 counted
CAD qty 11 each

24 MCMASTER 47065T245



overall 41.1 × 41.1 × 41.1 mm
P/N MCMASTER 47065T245 · Generic · 4 bodies
on hand unknown
intended configuration confirmed: no

8 this build 8 counted
CAD qty 8 each

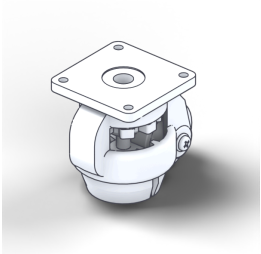
25 SLLB Enclosure 96in 47065T808

T-Slotted Framing

overall 2438 × 38.1 × 38.1 mm
Generic
on hand unknown
intended configuration confirmed: no

4 this build 4 counted
CAD qty 4 each

26 HOLKIE LEVELING CASTER



overall 104 × 86.2 × 73.5 mm (4.094 × 3.395 × 2.893 in)
P/N HOLKIE LEVELING CASTER · Generic · 10 bodies
on hand unknown
intended configuration confirmed: no

4 this build 4 counted
CAD qty 4 each

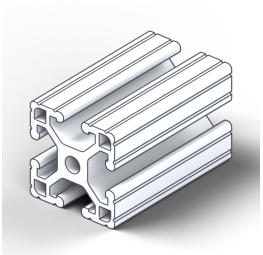
27 CASTER MOUNTING PLATE

overall 100 × 100 × 6.3 mm (3.953 × 3.953 × 0.25 in)
ALUMINUM, 6061-T651
on hand unknown
intended configuration confirmed: no

4 this build 4 counted
CAD qty 4 each

28 SLLB Enclosure 2.5in 47065T808

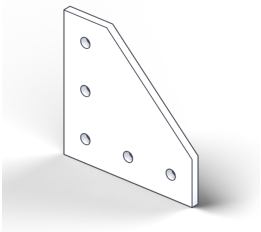
T-Slotted Framing



overall 63.5 × 38.1 × 38.1 mm
Generic
on hand unknown
intended configuration confirmed: no

2 this build 2 counted
CAD qty 2 each

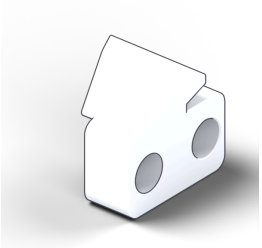
29 47065T271 ALUMINUM T-SLOTTED FRAMING EXTRUSION



overall 114 × 114 × 6.3 mm
Aluminum
on hand unknown
intended configuration confirmed: no

8 this build 8 counted
CAD qty 8 each

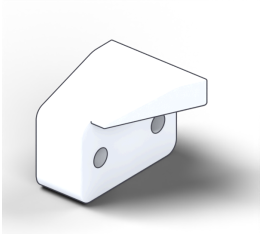
30 PANEL LOWER SUPPORT LEFT



overall 63.5 × 52.7 × 51.1 mm (2.5 × 2.075 × 2.01 in)
Generic
on hand unknown
intended configuration confirmed: no

4 this build 4 counted
CAD qty 4 each

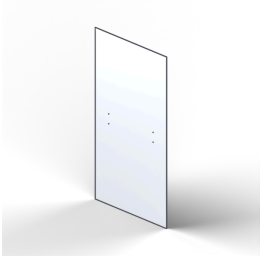
31 PANEL LOWER SUPPORT RIGHT



overall 63.5 × 52.7 × 51.1 mm (2.5 × 2.075 × 2.01 in)
ABS, BLACK
on hand unknown
intended configuration confirmed: no

4 this build 4 counted
CAD qty 4 each

32 ENCLOSURE PANEL

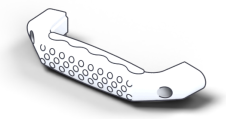


overall 2438 × 1143 × 3.2 mm (96 × 45 × 0.125 in)
ALUMINUM COMPOSITE MATERIAL
on hand unknown
intended configuration confirmed: no

4 this build 4 counted
CAD qty 4 each

33 MCMASTER 1950A67

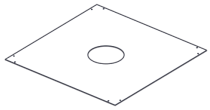
Unthreaded-Hole Pull Handle



overall 170 × 51.1 × 31.2 mm (6.71 × 2.014 × 1.23 in)
P/N MCMASTER_1950A67_Unthreaded-Hole Pull Handle · Generic
on hand unknown
intended configuration confirmed: no

8 this build 8 counted
CAD qty 8 each

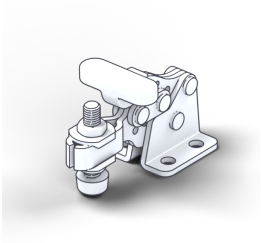
34 ENCLOSURE TOP PANEL



overall 1194 × 1194 × 3.2 mm (47 × 47 × 0.125 in)
P/N ENCLOSURE TOP PANEL · ALUMINUM COMPOSITE MATERIAL
on hand unknown
intended configuration confirmed: no

1 this build 1 counted ☐
CAD qty 1 each

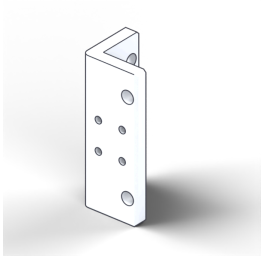
35 MCMASTER 5004A11 Compact Hold-Down Toggle Clamp



overall 56.5 × 35.7 × 28.4 mm (2.224 × 1.407 × 1.12 in)
P/N 5004A11_Compact Hold-Down Toggle Clamp · Generic · 20 bodies
on hand unknown
intended configuration confirmed: no

8 this build 8 counted ☐
CAD qty 8 each

36 TOGGLE CLAMP MOUNT



overall 76.2 × 30.5 × 30.5 mm (3 × 1.2 × 1.2 in)
ABS, BLACK
on hand unknown
intended configuration confirmed: no

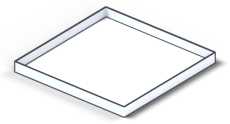
4 this build 4 counted ☐
CAD qty 4 each

The counted box is for a paper count only; recorded stock is on_hand_each in the project's manifests.

Deferred (not in this build)

These parts stay in the CAD model and its counts but are not part of phase-1: Enclosure without tanks and backlight.

37 SHULZE 2W.353 TANK



overall 991 × 991 × 82.6 mm (39 × 39 × 3.25 in)
Acrylic, Clear
on hand unknown
intended configuration confirmed: no
⚠ 1 coincident instance pair(s) in CAD: review before counting as stock

0 this build 0 counted ☐
CAD qty 3 each · deferred 3

Verification checklist: phase-1

Enclosure without tanks and backlight (draft, decided by Peter, 2026-10-22). Tick what you have for this build; quantities are CAD instance counts for the phase, not purchase quantities. Deferred parts are listed last for completeness.

#		Part	this build	cond.	counted	by	date	location	missing
1		MCMASTER 5537T293 T-Slotted Framing (CAD 2)	**2**	—					
2		MCMASTER 5537T429 T-Slotted Framing (CAD 2)	**2**	—					
3		SLLB Enclosure 15in 47065T808 T-Slotted Framing (CAD 2)	**2**	—					
4		SLLB Enclosure 22in 47065T808 T-Slotted Framing (CAD 2)	**2**	—					
5		SLLB Enclosure 41in 47065T808 T-Slotted Framing (CAD 2)	**2**	—					
6		MCMASTER 3136N149 T-Slotted Framing (CAD 2)	**2**	—					
7		MCMASTER 47065T27 T-Slotted Framing (CAD 4)	**4**	—					
8		MCMASTER 9293K121 Constant-Force Spring (CAD 2)	**2**	—					
9		MCMASTER 47065T845 Silver Corner Bracket (CAD 4)	**4**	—					
10		MCMASTER 5972K164 Ball Bearing (CAD 4)	**4**	—					
11		MCMASTER 90265A164 18-8 Stainless Steel Shoulder Screw (CAD 2)	**2**	—					
12		SPRING ROLLER (CAD 2)	**2**	—					
13		ROLLER MOUNT PLATE (CAD 2)	**2**	—					
14		SPRING MOUNT PLATE (CAD 2)	**2**	—					
15		CAMERA MOUNT PLATE (CAD 1)	**1**	—					
16		CAMERA MOUNT PLATE - TELEDYNE KINETIX (CAD 1)	**1**	—					
17		TSLOT EXTRUSION 1X40.5 copy (CAD 8)	**0**	8					
18		T SLOT CORNER 47065T242 (CAD 8)	**0**	8					
19		MCMASTER 9378K24 Natural Rubber Vibration-Damping Sandwich Mount (CAD 8)	**0**	8					
20		MCMASTER 90313A107 18-8 Stainless Steel Oversized Washer (CAD 8)	**0**	8					
21		TANK SUPPORT PLATFORM (CAD 2)	**0**	2					
22		TANK SUPPORT BRACKET (CAD 8)	**0**	8					
23		SLLB Enclosure 44in 47065T808 T-Slotted Framing (CAD 11)	**11**	—					
24		MCMASTER 47065T245 (CAD 8)	**8**	—					
25		SLLB Enclosure 96in 47065T808 T-Slotted Framing (CAD 4)	**4**	—					
26		HOLKIE LEVELING CASTER (CAD 4)	**4**	—					
27		CASTER MOUNTING PLATE (CAD 4)	**4**	—					
28		SLLB Enclosure 2.5in 47065T808 T-Slotted Framing (CAD 2)	**2**	—					
29		47065T271 ALUMINUM T-SLOTTED FRAMING EXTRUSION (CAD 8)	**8**	—					
30		PANEL LOWER SUPPORT LEFT (CAD 4)	**4**	—					
31		PANEL LOWER SUPPORT RIGHT (CAD 4)	**4**	—					
32		ENCLOSURE PANEL (CAD 4)	**4**	—					
33		MCMASTER 1950A67 Unthreaded-Hole Pull Handle (CAD 8)	**8**	—					
34		ENCLOSURE TOP PANEL (CAD 1)	**1**	—					
35		MCMASTER 5004A11 Compact Hold-Down Toggle Clamp (CAD 8)	**8**	—					
36		TOGGLE CLAMP MOUNT (CAD 4)	**4**	—					
37		SHULZE 2W.353 TANK (CAD 3)	deferred	—					

Supplementary items (not modelled in CAD)

Supplementary items are hardware, consumables or purchased items that are not modelled in CAD. They are listed for review and are never added to CAD quantities; an unknown quantity stays unknown. This is not an approved bill of materials. Status: draft: rows converted from the project procurement sheet; candidate item families, not an approved BOM; quantities unknown unless stated; never added to CAD counts.

#	Item	Kind, status	Required	Stock	Included with	Config. confirmed	counted
S1	**Caster-to-mounting-plate fastening** `SUP-001` joint `caster_to_plate` attaches to: CASTER ASSEMBLY:1, CASTER ASSEMBLY:2, CASTER ASSEMBLY:3, CASTER ASSEMBLY:4	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 4 caster assemblies; attachment points per caster unverified	on hand unknown	—	no	☐
S2	**Caster-plate-to-frame fastening** `SUP-002` joint `caster_plate_to_frame` attaches to: FREE STANDING FRAME:1, CASTER ASSEMBLY:1, CASTER ASSEMBLY:2, CASTER ASSEMBLY:3, CASTER ASSEMBLY:4	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 4 plates; frame attachments per plate unverified	on hand unknown	—	no	☐
S3	**Pull-handle-to-panel fastening** `SUP-003` joint `handle_to_panel` attaches to: PANEL ASSEMBLY:1, PANEL ASSEMBLY:2, PANEL ASSEMBLY:3, PANEL ASSEMBLY:4	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 8 handles on 4 panels; fastening points per handle unverified	on hand unknown	—	no	☐
S4	**Toggle-clamp-to-mount fastening** `SUP-004` joint `clamp_to_mount` attaches to: TOGGLE CLAMP ASSEMBLY:1, TOGGLE CLAMP ASSEMBLY:2, TOGGLE CLAMP ASSEMBLY:3, TOGGLE CLAMP ASSEMBLY:4	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 8 clamps on 4 mounts; points per clamp unverified	on hand unknown	—	no	☐
S5	**Toggle-clamp-mount-to-frame fastening** `SUP-005` joint `clamp_mount_to_frame` attaches to: TOGGLE CLAMP ASSEMBLY:1, TOGGLE CLAMP ASSEMBLY:2, TOGGLE CLAMP ASSEMBLY:3, TOGGLE CLAMP ASSEMBLY:4	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 4 mounts; points per mount unverified	on hand unknown	—	no	☐
S6	**Panel-lower-support fastening** `SUP-006` joint `lower_support_to_frame` attaches to: FREE STANDING FRAME:1	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 4 left + 4 right supports; points per support unverified	on hand unknown	—	no	☐
S7	**Main-frame connector hardware and joint completeness** `SUP-007` joint `main_frame_joints` attaches to: FREE STANDING FRAME:1	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 11 long rails + 4 columns + 2 short rails; joint count requires review	on hand unknown	—	no	☐
S8	**Camera-frame connector and bracket hardware**	item family, no SKU	required unknown (family,	on hand unknown	—	no	☐

#	Item	Kind, status	Required	Stock	Included with	Config. confirmed	counted
	SUP-008 joint camera_frame_joints attaches to: CAMERA FRAME:1	candidate, requires review	not specified); basis: 4 named 47065T845 brackets plus other connector/ sliding parts				
S9	**Tank-support-frame corner hardware** SUP-009 joint support_frame_corners attaches to: TANK SUPPORT FRAME:1, TANK SUPPORT FRAME:2	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 8 modeled corner connectors across 2 support frames	on hand unknown	—	no	☐
S10	**Tank-support-bracket attachment hardware** SUP-010 joint support_brackets attaches to: TANK SUPPORT FRAME:1, TANK SUPPORT FRAME:2	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 8 brackets across 2 support frames	on hand unknown	—	no	☐
S11	**Vibration-mount retention hardware** SUP-011 joint vibration_mounts attaches to: TANK SUPPORT FRAME:1, TANK SUPPORT FRAME:2	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 8 mounts + 8 oversized washers modeled	on hand unknown	—	no	☐
S12	**Camera, spring and roller plate attachment hardware** SUP-012 joint camera_plates attaches to: CAMERA FRAME:1	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 2 roller plates + 2 spring plates + 2 different camera plates	on hand unknown	—	no	☐
S13	**Spring end and roller/bearing retention hardware** SUP-013 joint spring_roller attaches to: CAMERA FRAME:1	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 2 springs + 2 rollers + 4 bearings + 2 shoulder screws modeled	on hand unknown	—	no	☐
S14	**Camera-to-mount attachment** SUP-014 joint camera_attachment attaches to: CAMERA FRAME:1	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: Camera plate variants present; actual camera selection unconfirmed	on hand unknown	—	no	☐
S15	**Top-panel retention and edge/ support interface** SUP-015 joint top_panel attaches to: ENCLOSURE TOP PANEL:1	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 1 enclosure top panel	on hand unknown	—	no	☐
S16	**Removable-panel retention interface** SUP-016 joint panel_retention attaches to: FREE STANDING FRAME:1, PANEL ASSEMBLY:1, PANEL ASSEMBLY:2, PANEL ASSEMBLY:3, PANEL ASSEMBLY:4, TOGGLE CLAMP ASSEMBLY:1, TOGGLE CLAMP	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 4 panels, 8 lower supports, 8 toggle clamps	on hand unknown	—	no	☐

#	Item	Kind, status	Required	Stock	Included with	Config. confirmed	counted
	ASSEMBLY:2, TOGGLE CLAMP ASSEMBLY:3, TOGGLE CLAMP ASSEMBLY:4						

4 new-design row(s) are in the appendix (not printed in this compact document).

Subassembly 1: CASTER ASSEMBLY (4 in this project)

shown: `FREE STANDING FRAME:1/CASTER ASSEMBLY:1` under `FREE STANDING FRAME:1` · other instances: `FREE STANDING FRAME:1/CASTER ASSEMBLY:2`, `FREE STANDING FRAME:1/CASTER ASSEMBLY:3`, `FREE STANDING FRAME:1/CASTER ASSEMBLY:4` · definition from the exporter's document

inspection view: occurrences at their CAD poses; not an assembly step; no sequence, fasteners or tool access implied.

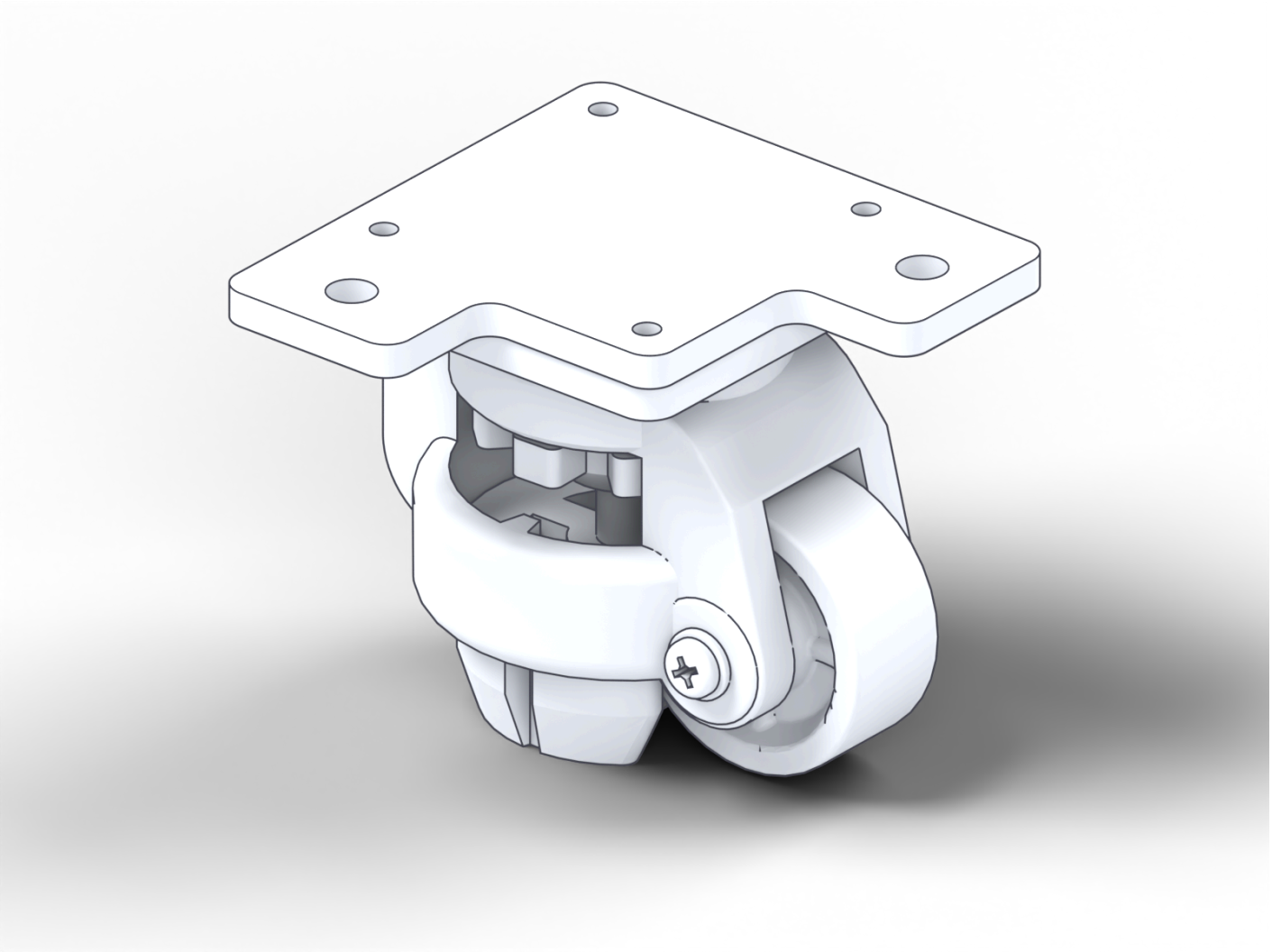


Figure 2: Subassembly 1: CASTER ASSEMBLY, instance `FREE STANDING FRAME:1/CASTER ASSEMBLY:1`, parts at their CAD positions (inspection view).

#	Picture	Part	Per subassembly	For all 4
26		HOLKIE LEVELING CASTER	×1	×4
27		CASTER MOUNTING PLATE	×1	×4

4 identical instances of this subassembly in the CAD model; the last column is the CAD count for all 4 (an instance count, not a purchase quantity). 2 occurrence(s) in the shown instance; ids in manual.json.

Hardware to review at this joint (not an approved BOM)

- S1 Caster-to-mounting-plate fastening · candidate, requires review · required unknown (family, not specified); basis: 4 caster assemblies; attachment points per caster unverified · joint `caster_to_plate`
- S2 Caster-plate-to-frame fastening · candidate, requires review · required unknown (family, not specified); basis: 4 plates; frame attachments per plate unverified · joint `caster_plate_to_frame`

Subassembly 2: PANEL ASSEMBLY (4 in this project)

shown: `PANEL ASSEMBLY:1` under `root` · other instances: `PANEL ASSEMBLY:2`, `PANEL ASSEMBLY:3`, `PANEL ASSEMBLY:4` · definition from the exporter's document

inspection view: occurrences at their CAD poses; not an assembly step; no sequence, fasteners or tool access implied.

Figure 3: Subassembly 2: PANEL ASSEMBLY, instance `PANEL ASSEMBLY:1`, parts at their CAD positions (inspection view).

#	Picture	Part	Per subassembly	For all 4
32		ENCLOSURE PANEL	×1	×4
33		MCMASTER 1950A67 Unthreaded-Hole Pull Handle	×2	×8

4 identical instances of this subassembly in the CAD model; the last column is the CAD count for all 4 (an instance count, not a purchase quantity). 3 occurrence(s) in the shown instance; ids in manual.json.

Hardware to review at this joint (not an approved BOM)

- S3 Pull-handle-to-panel fastening · candidate, requires review · required unknown (family, not specified); basis: 8 handles on 4 panels; fastening points per handle unverified · joint `handle_to_panel`
- S16 Removable-panel retention interface · candidate, requires review · required unknown (family, not specified); basis: 4 panels, 8 lower supports, 8 toggle clamps · joint `panel_retention`

Subassembly 3: TOGGLE CLAMP ASSEMBLY (4 in this project)

shown: `TOGGLE CLAMP ASSEMBLY:1` under `root` · other instances: `TOGGLE CLAMP ASSEMBLY:2`, `TOGGLE CLAMP ASSEMBLY:3`, `TOGGLE CLAMP ASSEMBLY:4` · definition from the exporter's document

inspection view: occurrences at their CAD poses; not an assembly step; no sequence, fasteners or tool access implied.

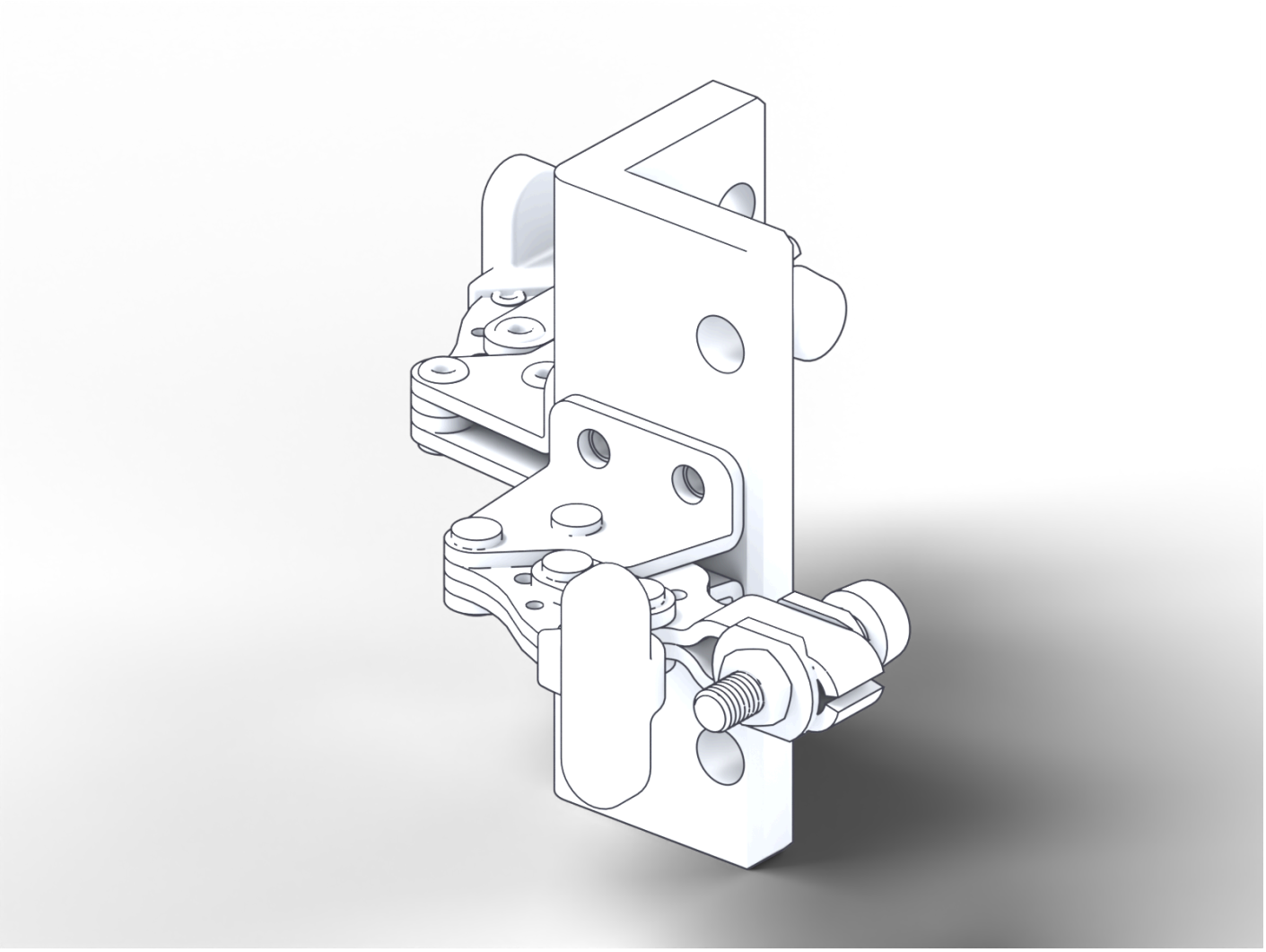
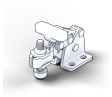


Figure 4: Subassembly 3: TOGGLE CLAMP ASSEMBLY, instance `TOGGLE CLAMP ASSEMBLY:1`, parts at their CAD positions (inspection view).

#	Picture	Part	Per subassembly	For all 4
35		MCMASTER 5004A11 Compact Hold-Down Toggle Clamp	×2	×8
36		TOGGLE CLAMP MOUNT	×1	×4

4 identical instances of this subassembly in the CAD model; the last column is the CAD count for all 4 (an instance count, not a purchase quantity). 3 occurrence(s) in the shown instance; ids in manual.json.

Hardware to review at this joint (not an approved BOM)

- S4 Toggle-clamp-to-mount fastening · candidate, requires review · required unknown (family, not specified); basis: 8 clamps on 4 mounts; points per clamp unverified · joint `clamp_to_mount`
- S5 Toggle-clamp-mount-to-frame fastening · candidate, requires review · required unknown (family, not specified); basis: 4 mounts; points per mount unverified · joint `clamp_mount_to_frame`

- **S16** Removable-panel retention interface · candidate, requires review · required unknown (family, not specified); basis: 4 panels, 8 lower supports, 8 toggle clamps · joint `panel_retention`

Subassembly 4: TANK SUPPORT FRAME (2 in this project)

shown: `TANK SUPPORT FRAME:1` under `root` · other instance: `TANK SUPPORT FRAME:2` · definition from the exporter's document

inspection view: occurrences at their CAD poses; not an assembly step; no sequence, fasteners or tool access implied.

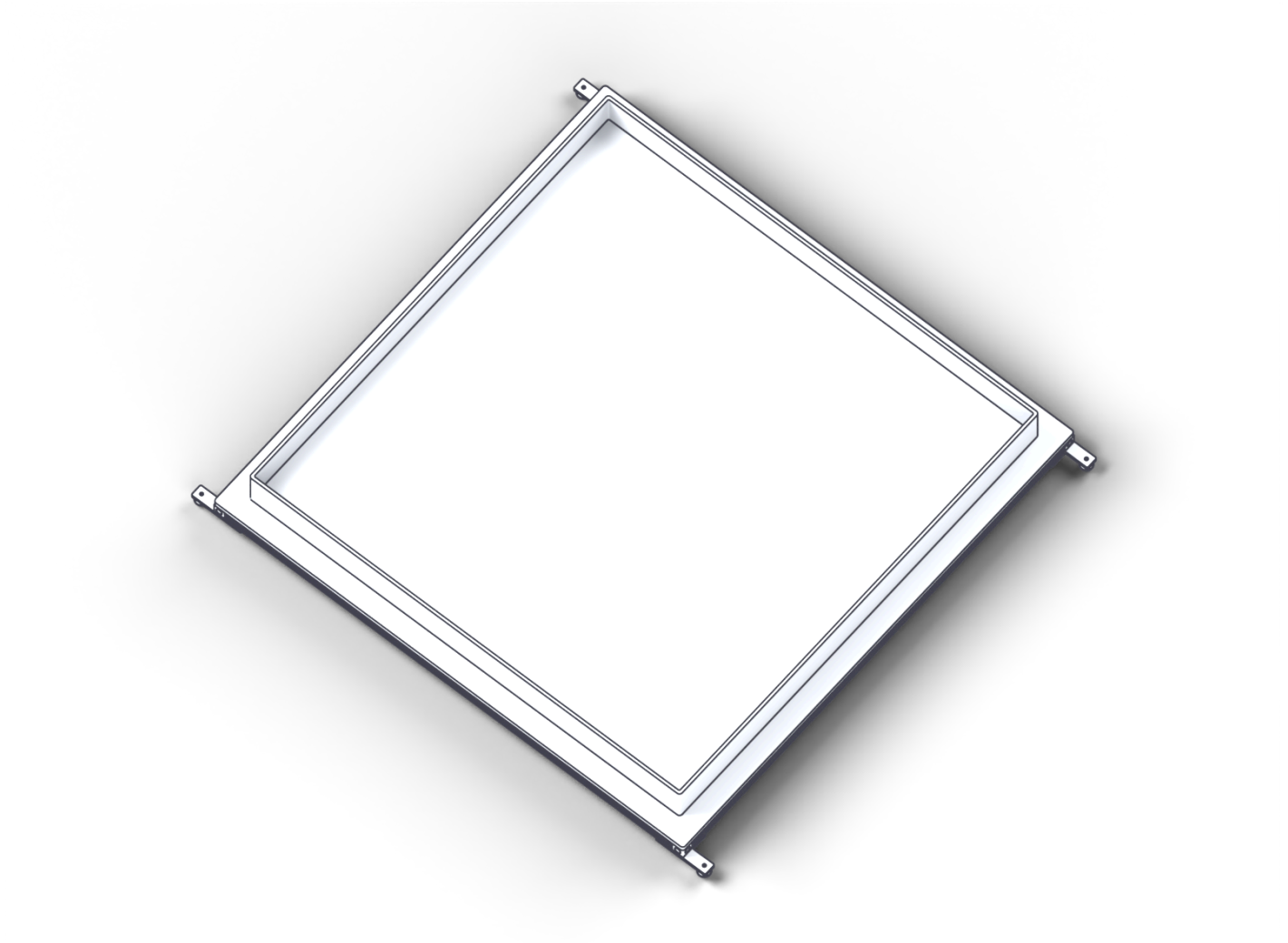
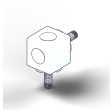
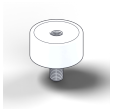


Figure 5: Subassembly 4: TANK SUPPORT FRAME, instance `TANK SUPPORT FRAME:1`, parts at their CAD positions (inspection view).

#	Picture	Part	Per subassembly	For all 2
17		TSLOT EXTRUSION 1X40.5 copy	×4	×8
18		T SLOT CORNER 47065T242	×4	×8
19		MCMASTER 9378K24 Natural Rubber Vibration-Damping Sandwich Mount	×4	×8
20		MCMASTER 90313A107 18-8 Stainless Steel Oversized Washer	×4	×8

#	Picture	Part	Per subassembly	For all 2
21		TANK SUPPORT PLATFORM	×1	×2
22		TANK SUPPORT BRACKET	×4	×8
37		SHULZE 2W.353 TANK	×1	×2

2 identical instances of this subassembly in the CAD model; the last column is the CAD count for all 2 (an instance count, not a purchase quantity). 22 occurrence(s) in the shown instance; ids in manual.json.

Hardware to review at this joint (not an approved BOM)

- S9 Tank-support-frame corner hardware · candidate, requires review · required unknown (family, not specified); basis: 8 modeled corner connectors across 2 support frames · joint `support_frame_corners`
- S10 Tank-support-bracket attachment hardware · candidate, requires review · required unknown (family, not specified); basis: 8 brackets across 2 support frames · joint `support_brackets`
- S11 Vibration-mount retention hardware · candidate, requires review · required unknown (family, not specified); basis: 8 mounts + 8 oversized washers modeled · joint `vibration_mounts`

Subassembly 5: CAMERA FRAME (1 in this project)

shown: `CAMERA FRAME:1` under `root` · definition from the exporter's document · flexible assembly: positions may differ per instance; 32 position(s) overridden against the definition

inspection view: occurrences at their CAD poses; not an assembly step; no sequence, fasteners or tool access implied.

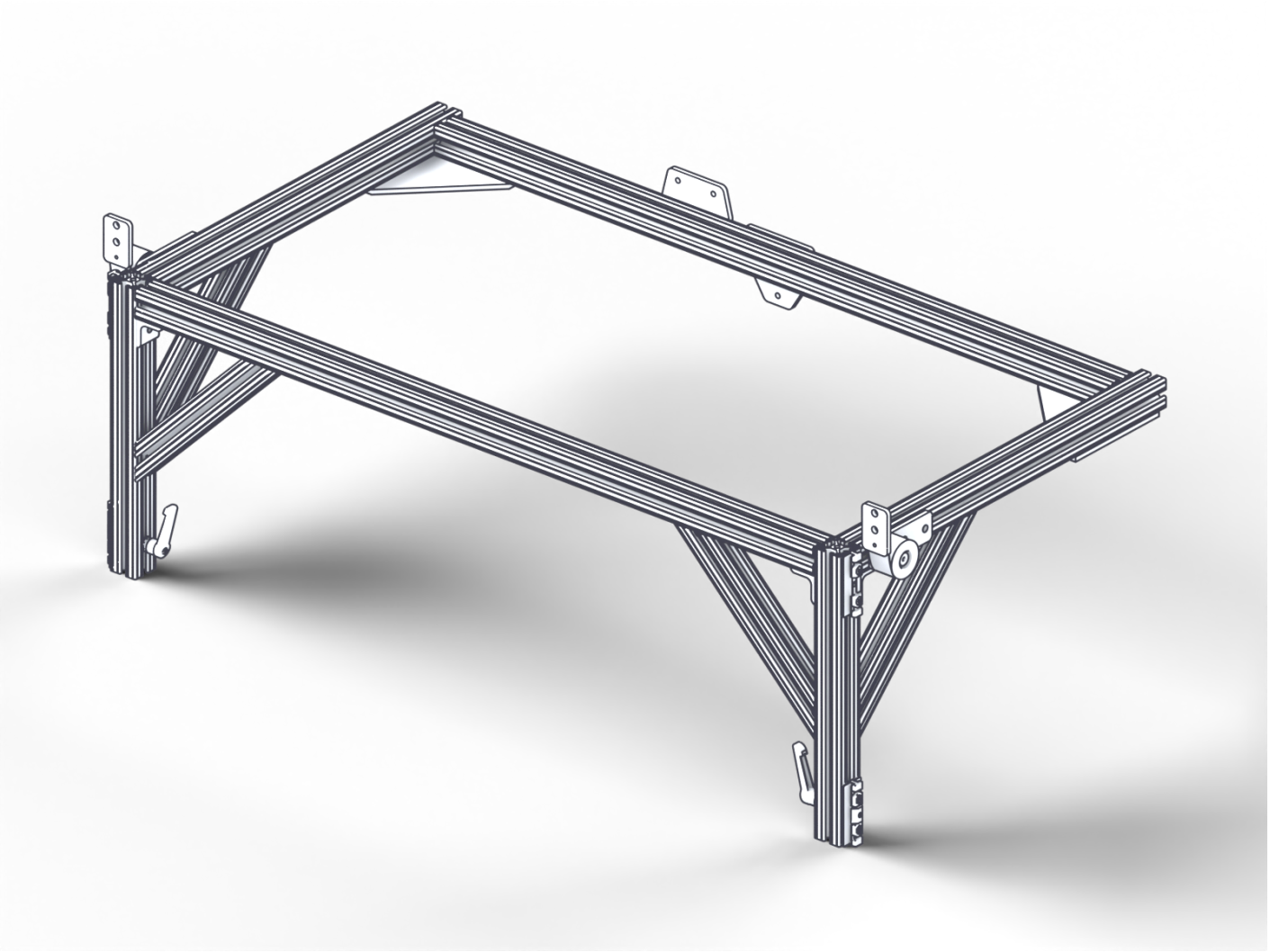
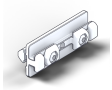
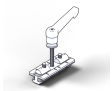
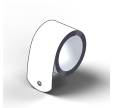


Figure 6: Subassembly 5: CAMERA FRAME, instance `CAMERA FRAME:1`, parts at their CAD positions (inspection view).

#	Picture	Part	In this subassembly
1		MCMASTER 5537T293 T-Slotted Framing	×2
2		MCMASTER 5537T429 T-Slotted Framing	×2
3		SLLB Enclosure 15in 47065T808 T-Slotted Framing	×2
4		SLLB Enclosure 22in 47065T808 T-Slotted Framing	×2
5		SLLB Enclosure 41in 47065T808 T-Slotted Framing	×2

#	Picture	Part	In this subassembly
6		MCMASTER 3136N149 T-Slotted Framing	×2
7		MCMASTER 47065T27 T-Slotted Framing	×4
8		MCMASTER 9293K121 Constant-Force Spring	×2
9		MCMASTER 47065T845 Silver Corner Bracket	×4
10		MCMASTER 5972K164 Ball Bearing	×4
11		MCMASTER 90265A164 18-8 Stainless Steel Shoulder Screw	×2
12		SPRING ROLLER	×2
13		ROLLER MOUNT PLATE	×2
14		SPRING MOUNT PLATE	×2
15		CAMERA MOUNT PLATE	×1
16		CAMERA MOUNT PLATE - TELEDYNE KINETIX	×1

36 occurrence(s) in the shown instance; ids in manual.json.

Hardware to review at this joint (not an approved BOM)

- S8 Camera-frame connector and bracket hardware · candidate, requires review · required unknown (family, not specified); basis: 4 named 47065T845 brackets plus other connector/sliding parts · joint `camera_frame_joints`
- S12 Camera, spring and roller plate attachment hardware · candidate, requires review · required unknown (family, not specified); basis: 2 roller plates + 2 spring plates + 2 different camera plates · joint `camera_plates`
- S13 Spring end and roller/bearing retention hardware · candidate, requires review · required unknown (family, not specified); basis: 2 springs + 2 rollers + 4 bearings + 2 shoulder screws modeled · joint `spring_roller`
- S14 Camera-to-mount attachment · candidate, requires review · required unknown (family, not specified); basis: Camera plate variants present; actual camera selection unconfirmed · joint `camera_attachment`

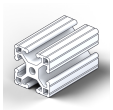
Subassembly 6: FREE STANDING FRAME (1 in this project)

shown: `FREE STANDING FRAME:1` under `root` · definition from the exporter's document · flexible assembly: positions may differ per instance; 12 position(s) overridden against the definition

inspection view: occurrences at their CAD poses; not an assembly step; no sequence, fasteners or tool access implied.

Figure 7: Subassembly 6: FREE STANDING FRAME, instance `FREE STANDING FRAME:1`, parts at their CAD positions (inspection view).

#	Picture	Part	In this subassembly
23		SLLB Enclosure 44in 47065T808 T-Slotted Framing	×11
24		MCMASTER 47065T245	×8
25		SLLB Enclosure 96in 47065T808 T-Slotted Framing	×4
26		HOLKIE LEVELING CASTER	×4

#	Picture	Part	In this subassembly
27		CASTER MOUNTING PLATE	×4
28		SLLB Enclosure 2.5in 47065T808 T-Slotted Framing	×2
29		47065T271 ALUMINUM T-SLOTTED FRAMING EXTRUSION	×8
30		PANEL LOWER SUPPORT LEFT	×4
31		PANEL LOWER SUPPORT RIGHT	×4

49 occurrence(s) in the shown instance; ids in manual.json.

Hardware to review at this joint (not an approved BOM)

- S1 Caster-to-mounting-plate fastening · candidate, requires review · required unknown (family, not specified); basis: 4 caster assemblies; attachment points per caster unverified · joint `caster_to_plate`
- S2 Caster-plate-to-frame fastening · candidate, requires review · required unknown (family, not specified); basis: 4 plates; frame attachments per plate unverified · joint `caster_plate_to_frame`
- S6 Panel-lower-support fastening · candidate, requires review · required unknown (family, not specified); basis: 4 left + 4 right supports; points per support unverified · joint `lower_support_to_frame`
- S7 Main-frame connector hardware and joint completeness · candidate, requires review · required unknown (family, not specified); basis: 11 long rails + 4 columns + 2 short rails; joint count requires review · joint `main_frame_joints`
- S16 Removable-panel retention interface · candidate, requires review · required unknown (family, not specified); basis: 4 panels, 8 lower supports, 8 toggle clamps · joint `panel_retention`

Joint 1: caster_to_plate

context `FREE STANDING FRAME:1/CASTER ASSEMBLY:1` (faded, 0 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

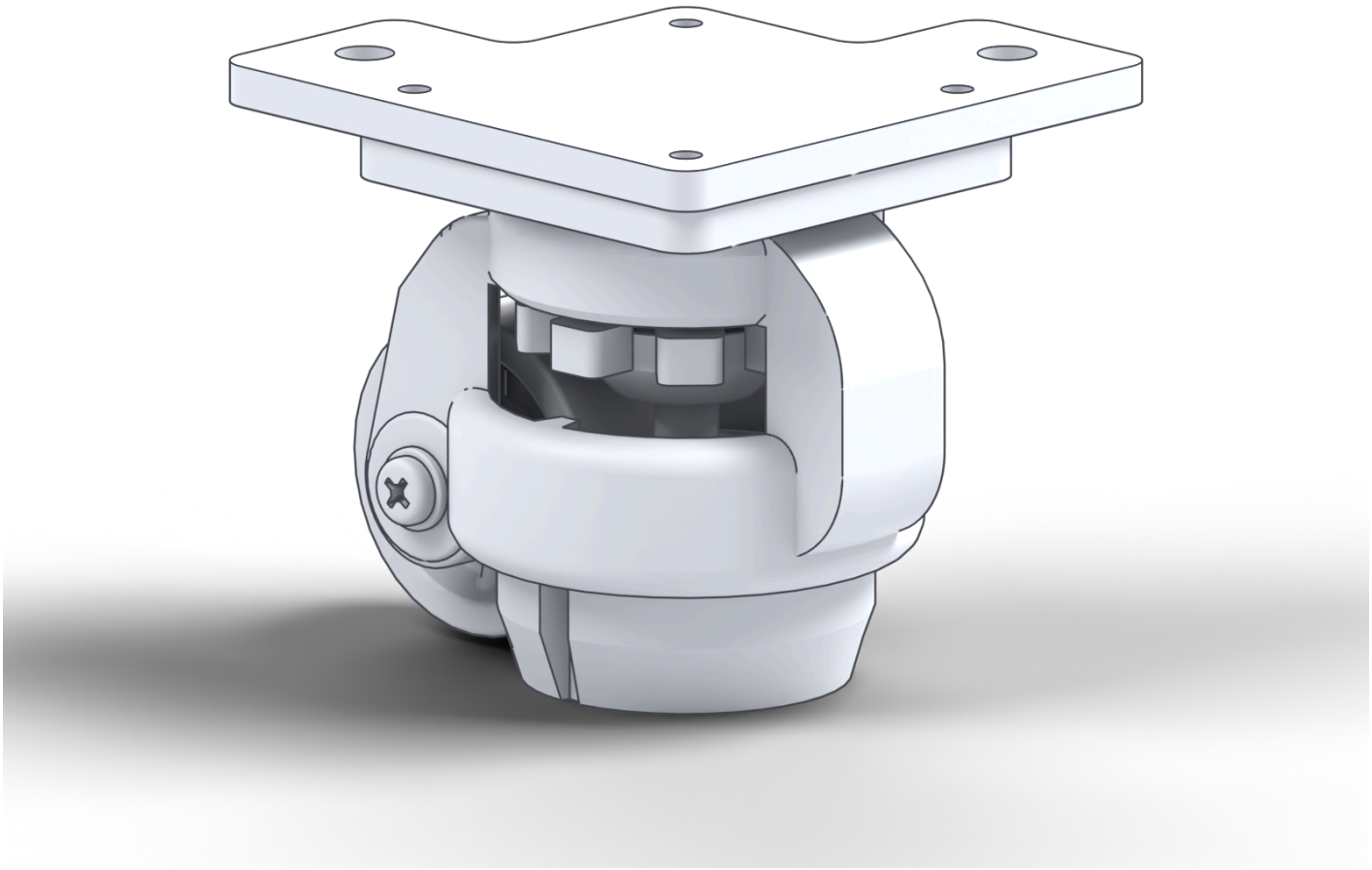
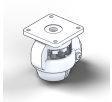


Figure 8: Joint 1: caster_to_plate, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
26		HOLKIE LEVELING CASTER `FREE STANDING FRAME:1/CASTER ASSEMBLY:1/HOLKIE LEVELING CASTER:1`
27		CASTER MOUNTING PLATE `FREE STANDING FRAME:1/CASTER ASSEMBLY:1/CASTER MOUNTING PLATE:1`

Hardware to review at this joint (not an approved BOM)

- S1 Caster-to-mounting-plate fastening · candidate, requires review · required unknown (family, not specified); basis: 4 caster assemblies; attachment points per caster unverified · joint `caster_to_plate`

Joint 2: caster_plate_to_frame

context `FREE STANDING FRAME:1` (faded, 48 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

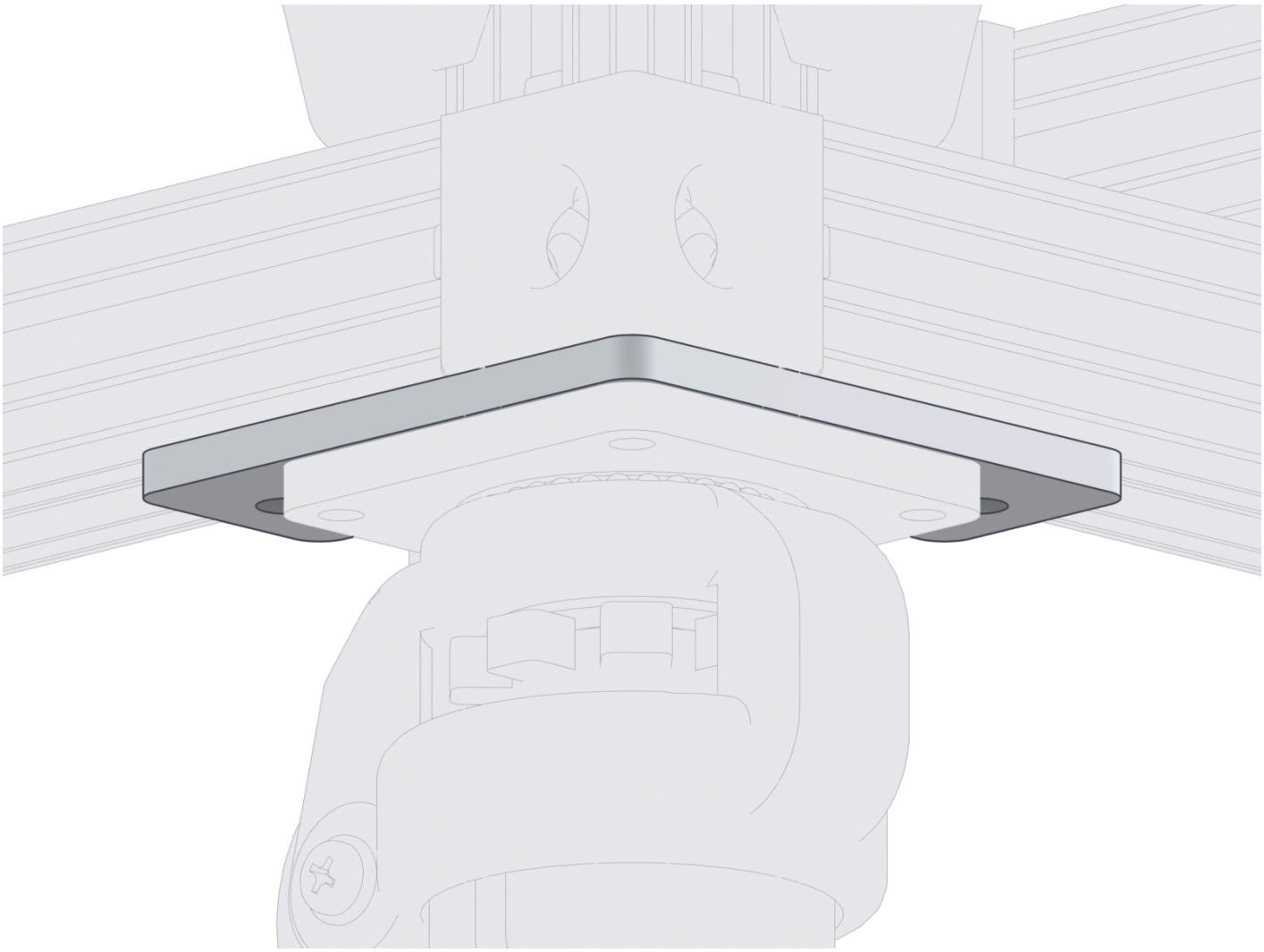


Figure 9: Joint 2: caster_plate_to_frame, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
27		CASTER MOUNTING PLATE `FREE STANDING FRAME:1/CASTER ASSEMBLY:1/CASTER MOUNTING PLATE:1`

Hardware to review at this joint (not an approved BOM)

- S2 Caster-plate-to-frame fastening · candidate, requires review · required unknown (family, not specified); basis: 4 plates; frame attachments per plate unverified · joint `caster_plate_to_frame`

Joint 3: handle_to_panel

context `PANEL ASSEMBLY:1` (faded, 2 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

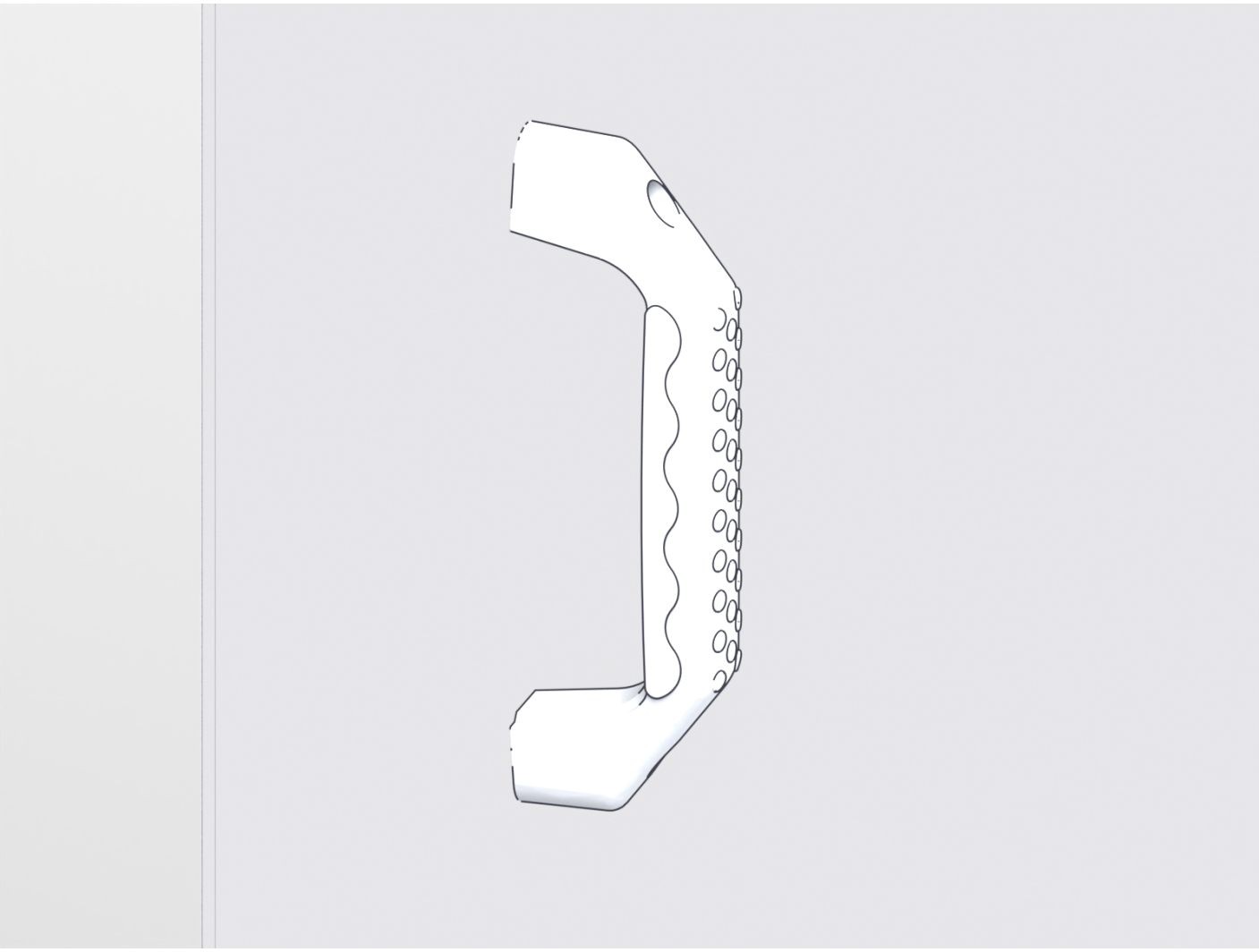


Figure 10: Joint 3: handle_to_panel, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
33		MCMASTER 1950A67 Unthreaded-Hole Pull Handle `PANEL ASSEMBLY:1/MCMASTER_1950A67_Unthreaded-Hole Pull Handle:1`

Hardware to review at this joint (not an approved BOM)

- S3 Pull-handle-to-panel fastening · candidate, requires review · required unknown (family, not specified); basis: 8 handles on 4 panels; fastening points per handle unverified · joint `handle_to_panel`

Joint 4: clamp_to_mount

context `TOGGLE CLAMP ASSEMBLY:1` (faded, 1 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

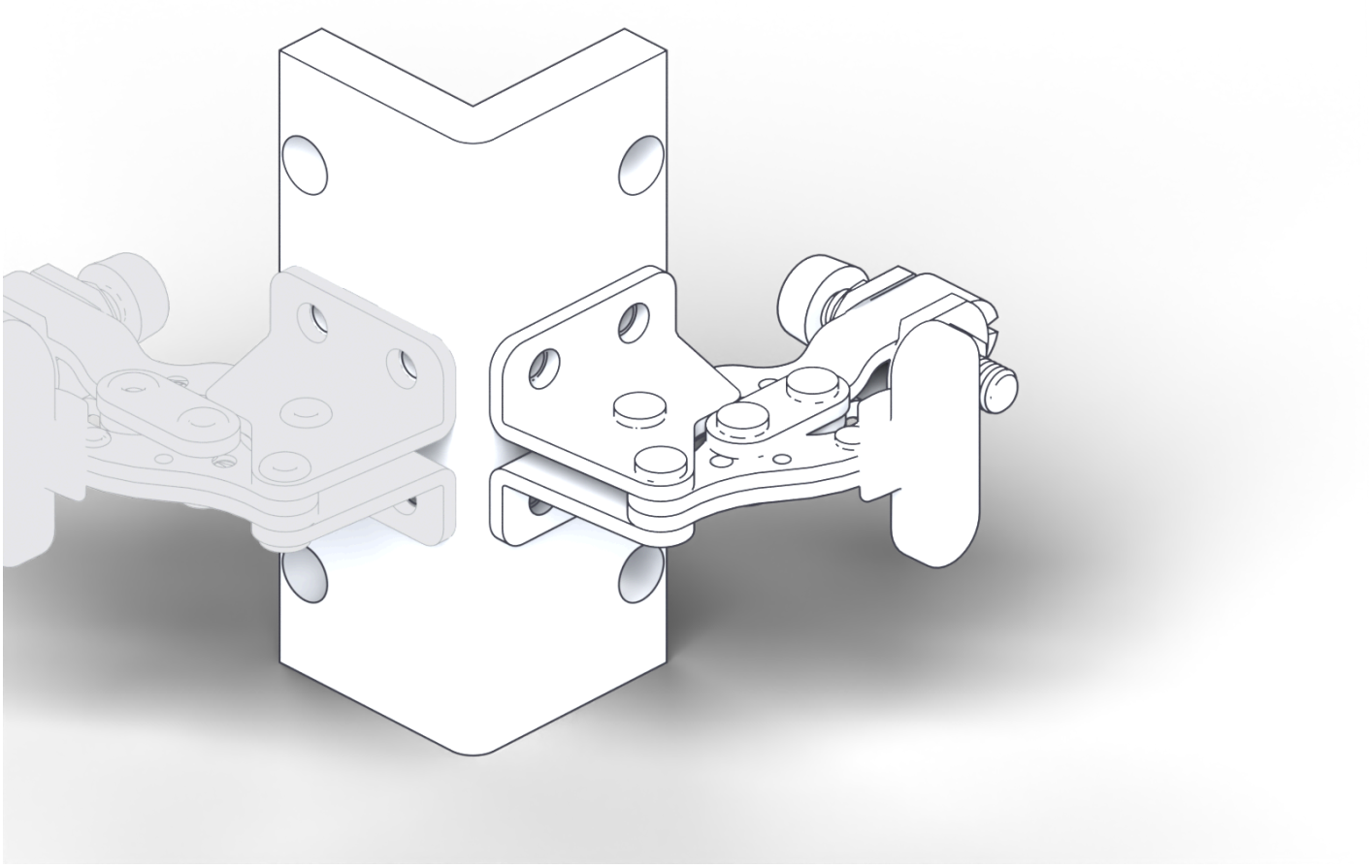
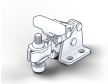


Figure 11: Joint 4: clamp_to_mount, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
35		MCMASTER 5004A11 Compact Hold-Down Toggle Clamp `TOGGLE CLAMP ASSEMBLY:1/MCMASTER 5004A11_Compact Hold-Down Toggle Clamp:1`
36		TOGGLE CLAMP MOUNT `TOGGLE CLAMP ASSEMBLY:1/TOGGLE CLAMP MOUNT:1`

Hardware to review at this joint (not an approved BOM)

- S4 Toggle-clamp-to-mount fastening · candidate, requires review · required unknown (family, not specified); basis: 8 clamps on 4 mounts; points per clamp unverified · joint `clamp_to_mount`

Joint 5: clamp_mount_to_frame

context `FREE STANDING FRAME:1` (faded, 49 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

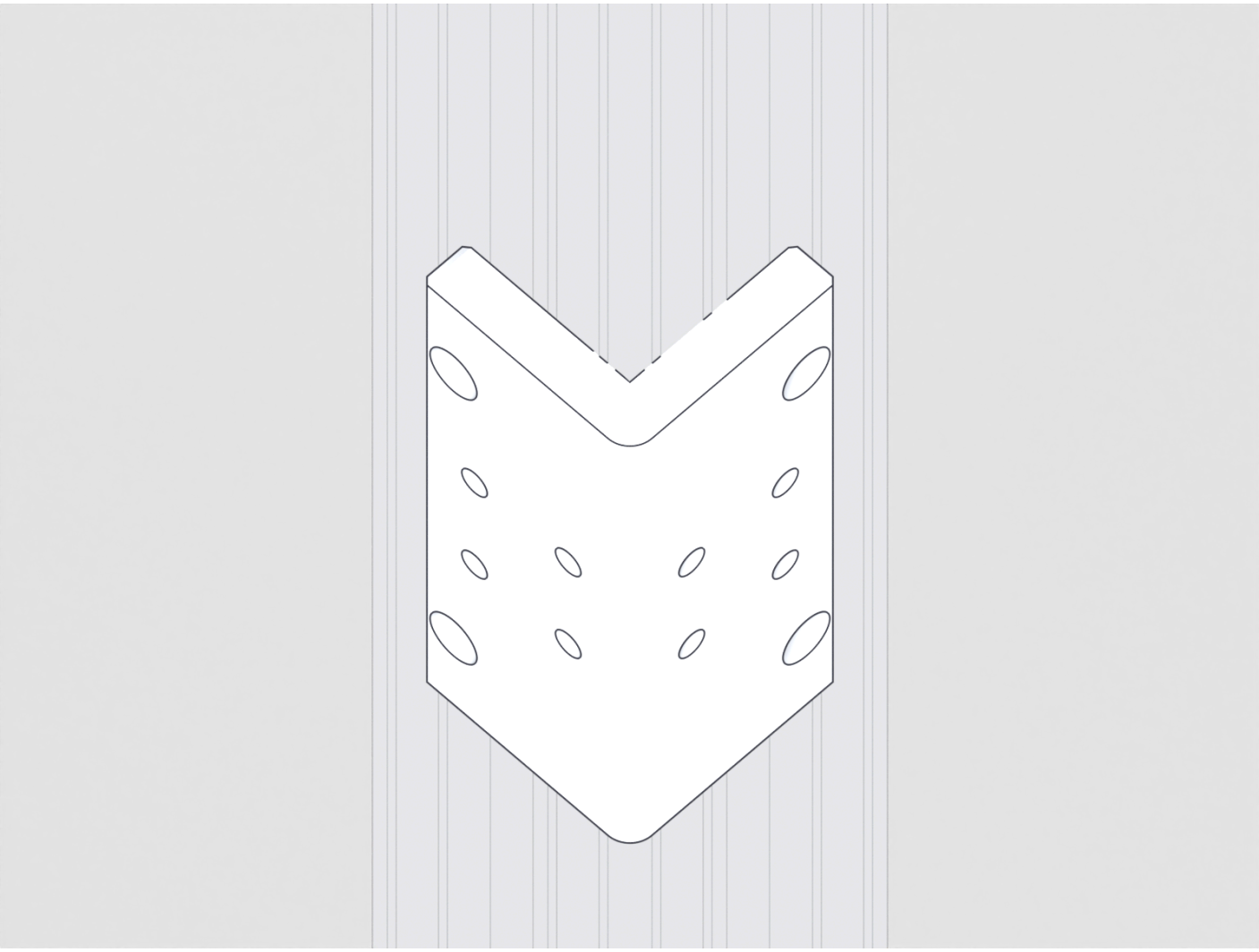
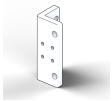


Figure 12: Joint 5: clamp_mount_to_frame, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
36		TOGGLE CLAMP MOUNT `TOGGLE CLAMP ASSEMBLY:1/TOGGLE CLAMP MOUNT:1`

Hardware to review at this joint (not an approved BOM)

- S5 Toggle-clamp-mount-to-frame fastening · candidate, requires review · required unknown (family, not specified); basis: 4 mounts; points per mount unverified · joint `clamp_mount_to_frame`

Joint 6: lower_support_to_frame

context `FREE STANDING FRAME:1` (faded, 48 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

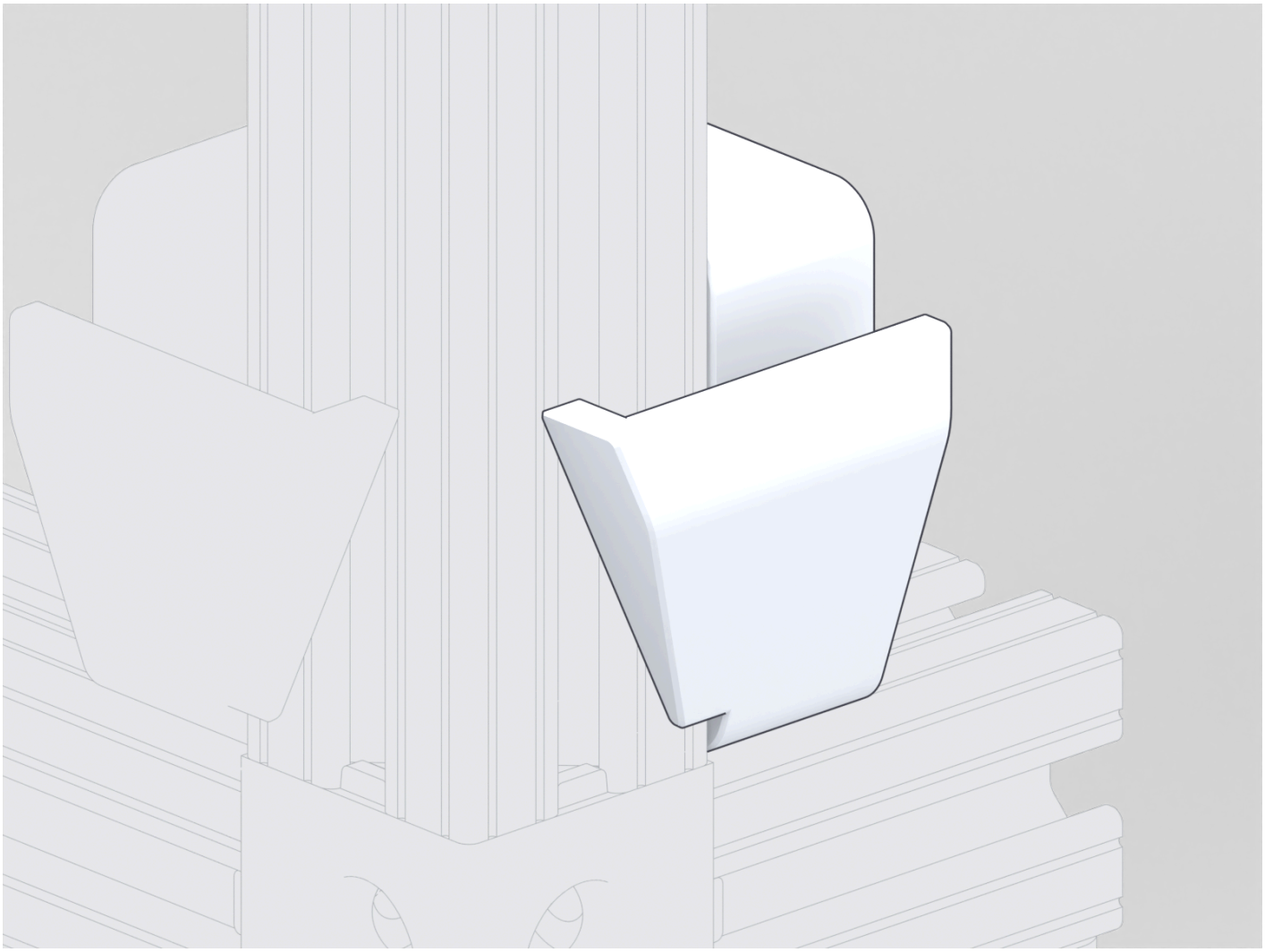


Figure 13: Joint 6: lower_support_to_frame, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
30		PANEL LOWER SUPPORT LEFT `FREE STANDING FRAME:1/PANEL LOWER SUPPORT LEFT:1`

Hardware to review at this joint (not an approved BOM)

- S6 Panel-lower-support fastening · candidate, requires review · required unknown (family, not specified); basis: 4 left + 4 right supports; points per support unverified · joint `lower_support_to_frame`

Joint 7: main_frame_joints

context `FREE STANDING FRAME:1` (faded, 48 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

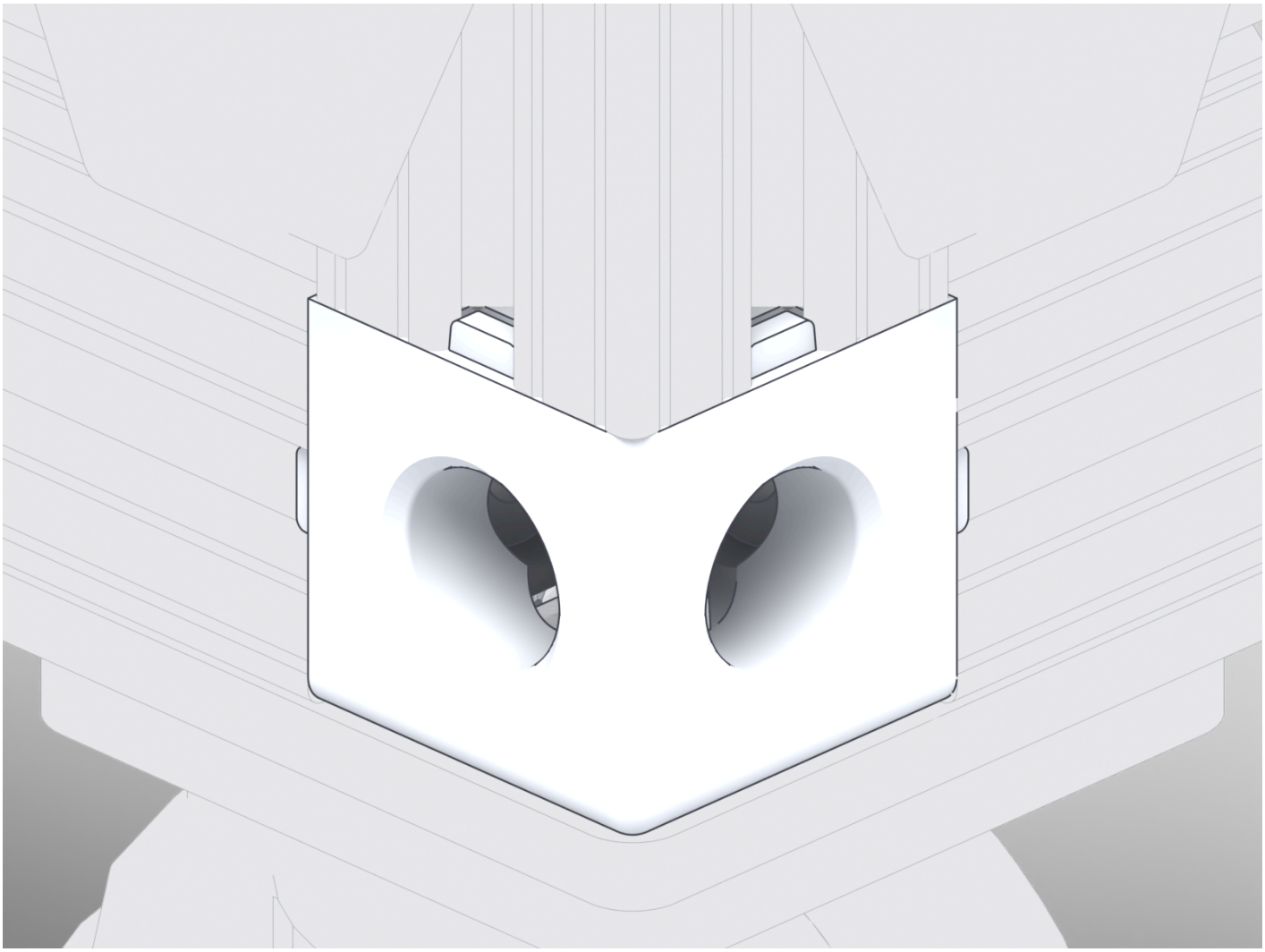


Figure 14: Joint 7: main_frame_joints, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
24		MCMASTER 47065T245 `FREE STANDING FRAME:1/MCMASTER 47065T245:1`

Hardware to review at this joint (not an approved BOM)

- S7 Main-frame connector hardware and joint completeness · candidate, requires review · required unknown (family, not specified); basis: 11 long rails + 4 columns + 2 short rails; joint count requires review · joint `main_frame_joints`

Joint 8: camera_frame_joints

context `CAMERA FRAME:1` (faded, 35 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

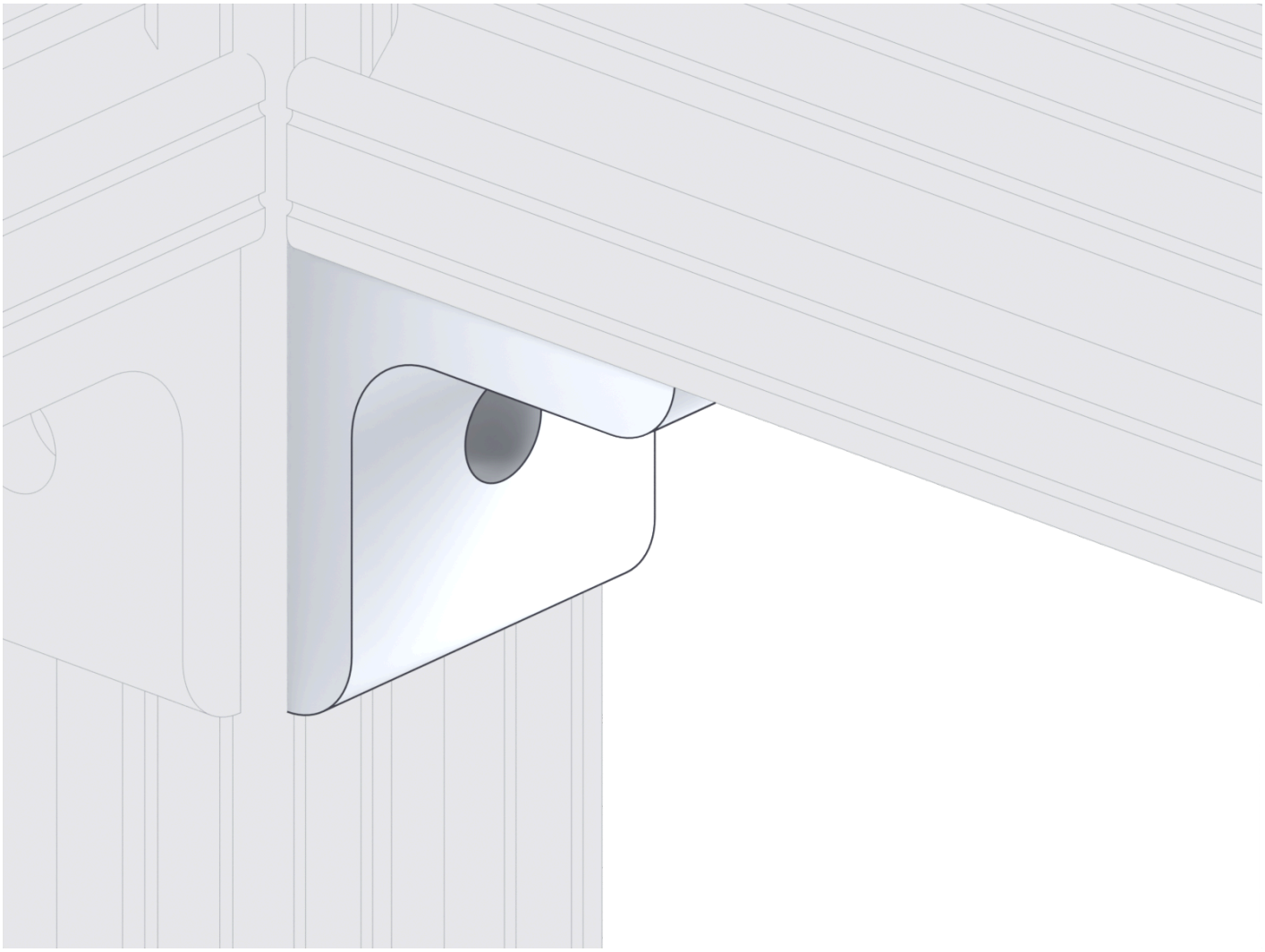


Figure 15: Joint 8: camera_frame_joints, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
9		MCMASTER 47065T845 Silver Corner Bracket `CAMERA FRAME:1/MCMASTER 47065T845_Silver Corner Bracket:1`

Hardware to review at this joint (not an approved BOM)

- S8 Camera-frame connector and bracket hardware · candidate, requires review · required unknown (family, not specified); basis: 4 named 47065T845 brackets plus other connector/sliding parts · joint `camera_frame_joints`

Joint 9: support_frame_corners

context `TANK SUPPORT FRAME:1` (faded, 21 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

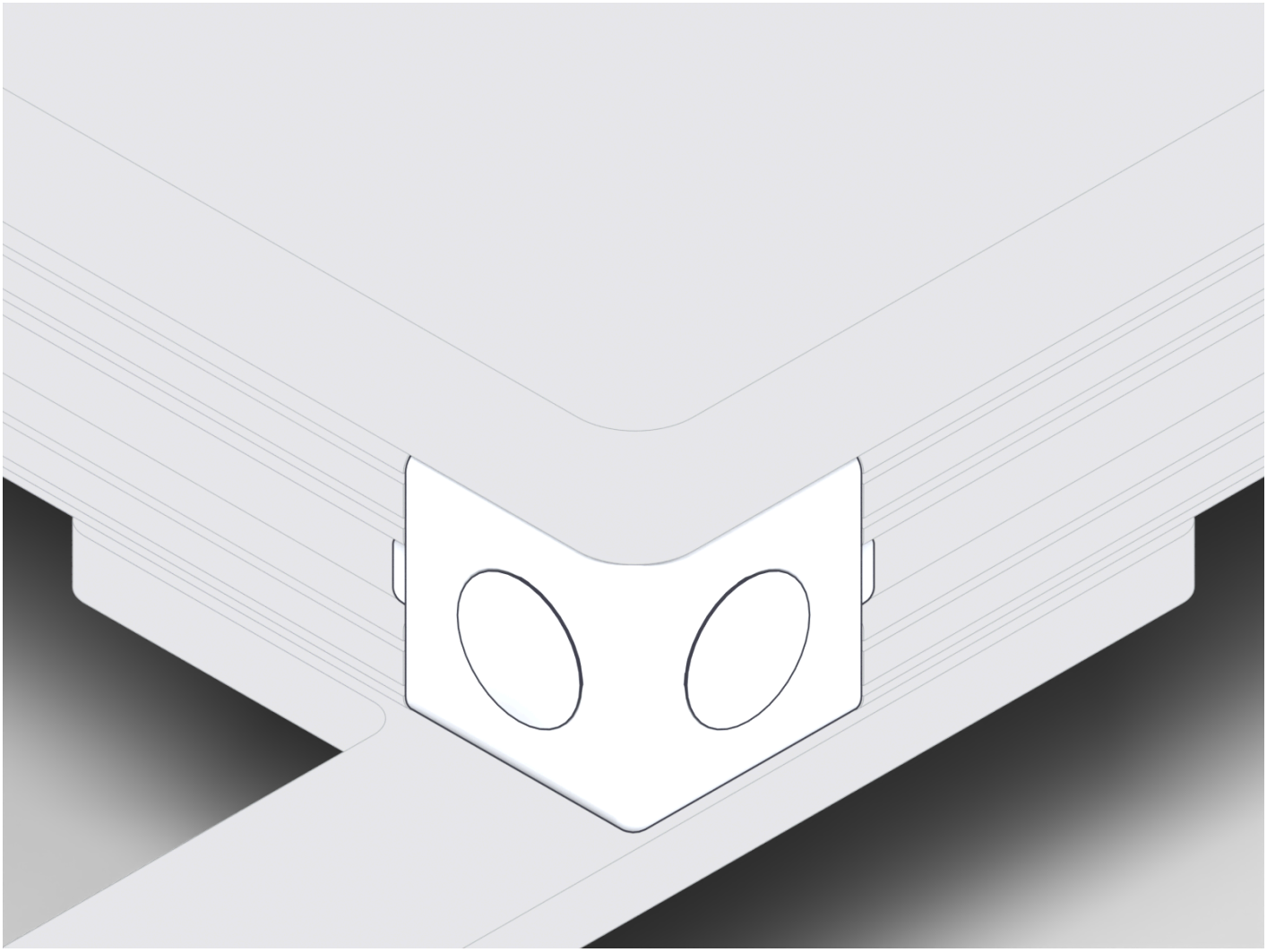
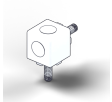


Figure 16: Joint 9: support_frame_corners, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
18		T SLOT CORNER 47065T242 `TANK SUPPORT FRAME:1/T SLOT CORNER 47065T242:1`

Hardware to review at this joint (not an approved BOM)

- S9 Tank-support-frame corner hardware · candidate, requires review · required unknown (family, not specified); basis: 8 modeled corner connectors across 2 support frames · joint `support_frame_corners`

Joint 10: support_brackets

context `TANK SUPPORT FRAME:1` (faded, 21 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

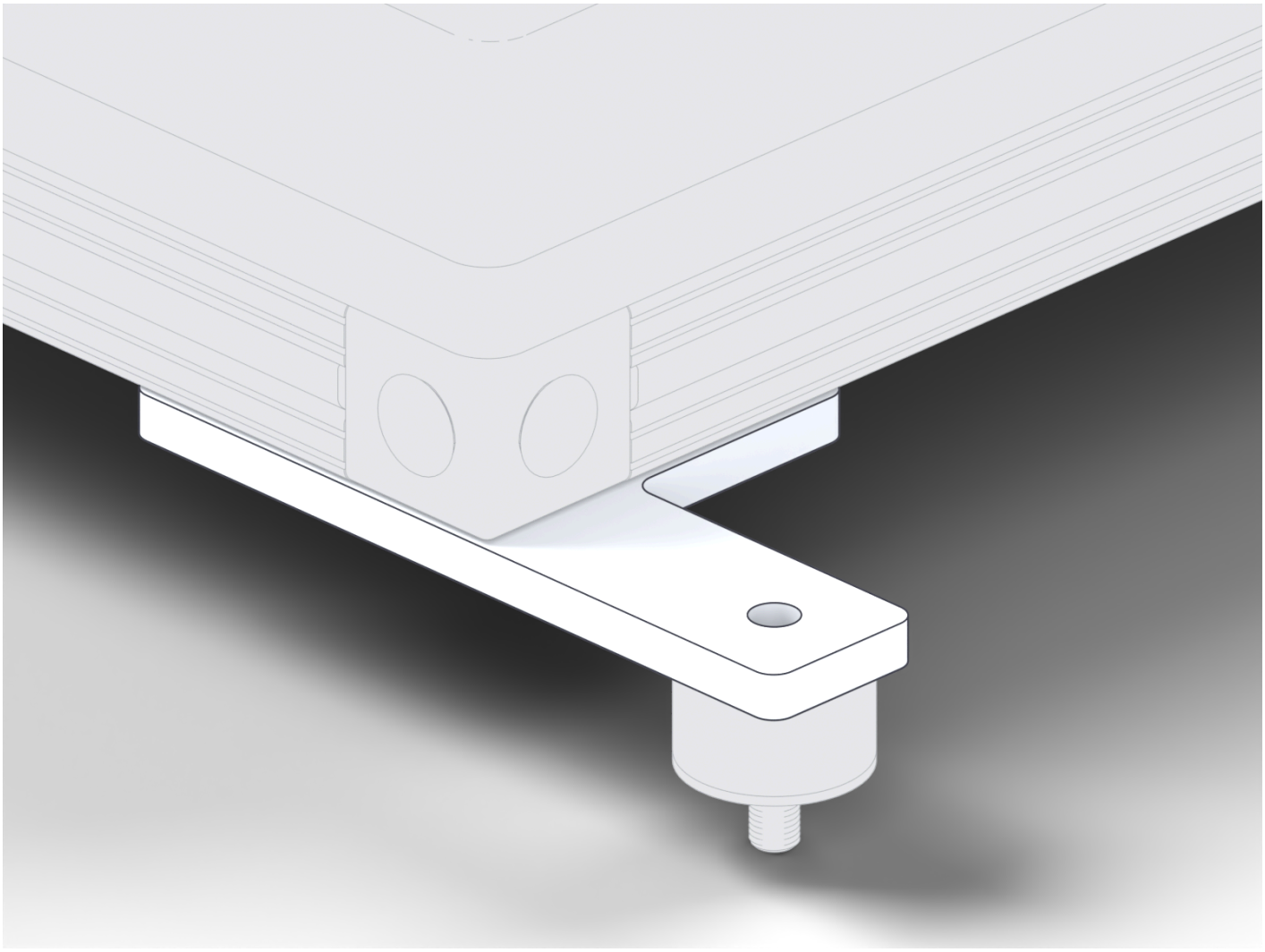


Figure 17: Joint 10: support_brackets, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
22		TANK SUPPORT BRACKET `TANK SUPPORT FRAME:1/TANK SUPPORT BRACKET:1`

Hardware to review at this joint (not an approved BOM)

- S10 Tank-support-bracket attachment hardware · candidate, requires review · required unknown (family, not specified); basis: 8 brackets across 2 support frames · joint `support_brackets`

Joint 11: vibration_mounts

context `TANK SUPPORT FRAME:1` (faded, 20 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

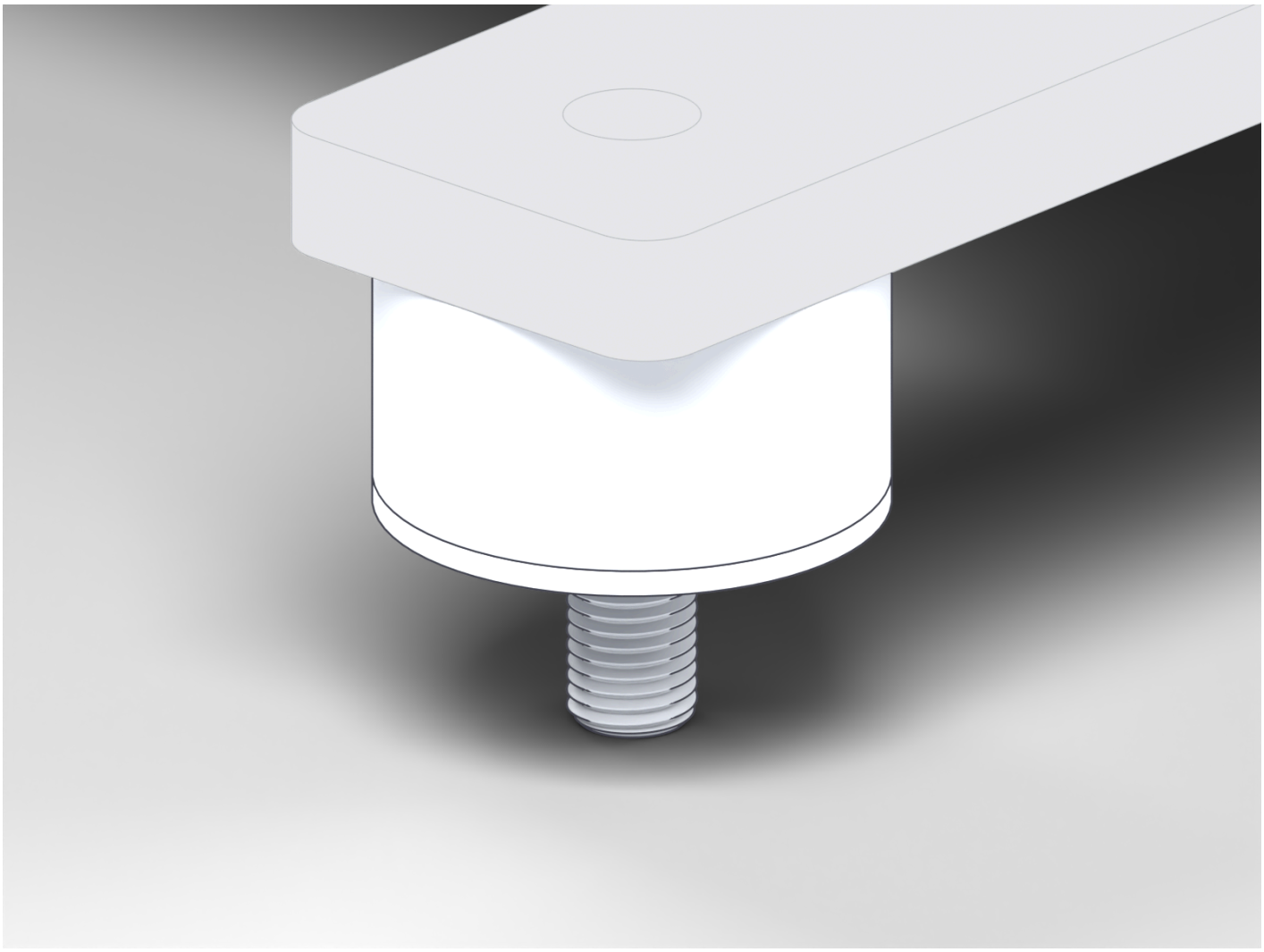
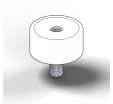


Figure 18: Joint 11: vibration_mounts, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
19		MCMASTER 9378K24 Natural Rubber Vibration-Damping Sandwich Mount `TANK SUPPORT FRAME:1/MCMASTER 9378K24_Natural Rubber Vibration-Damping Sandwich Mount:1`
20		MCMASTER 90313A107 18-8 Stainless Steel Oversized Washer `TANK SUPPORT FRAME:1/MCMASTER 90313A107_18-8 Stainless Steel Oversized Washer:4`

Hardware to review at this joint (not an approved BOM)

- S11 Vibration-mount retention hardware · candidate, requires review · required unknown (family, not specified); basis: 8 mounts + 8 oversized washers modeled · joint `vibration_mounts`

Joint 12: camera_plates

context `CAMERA FRAME:1` (faded, 34 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

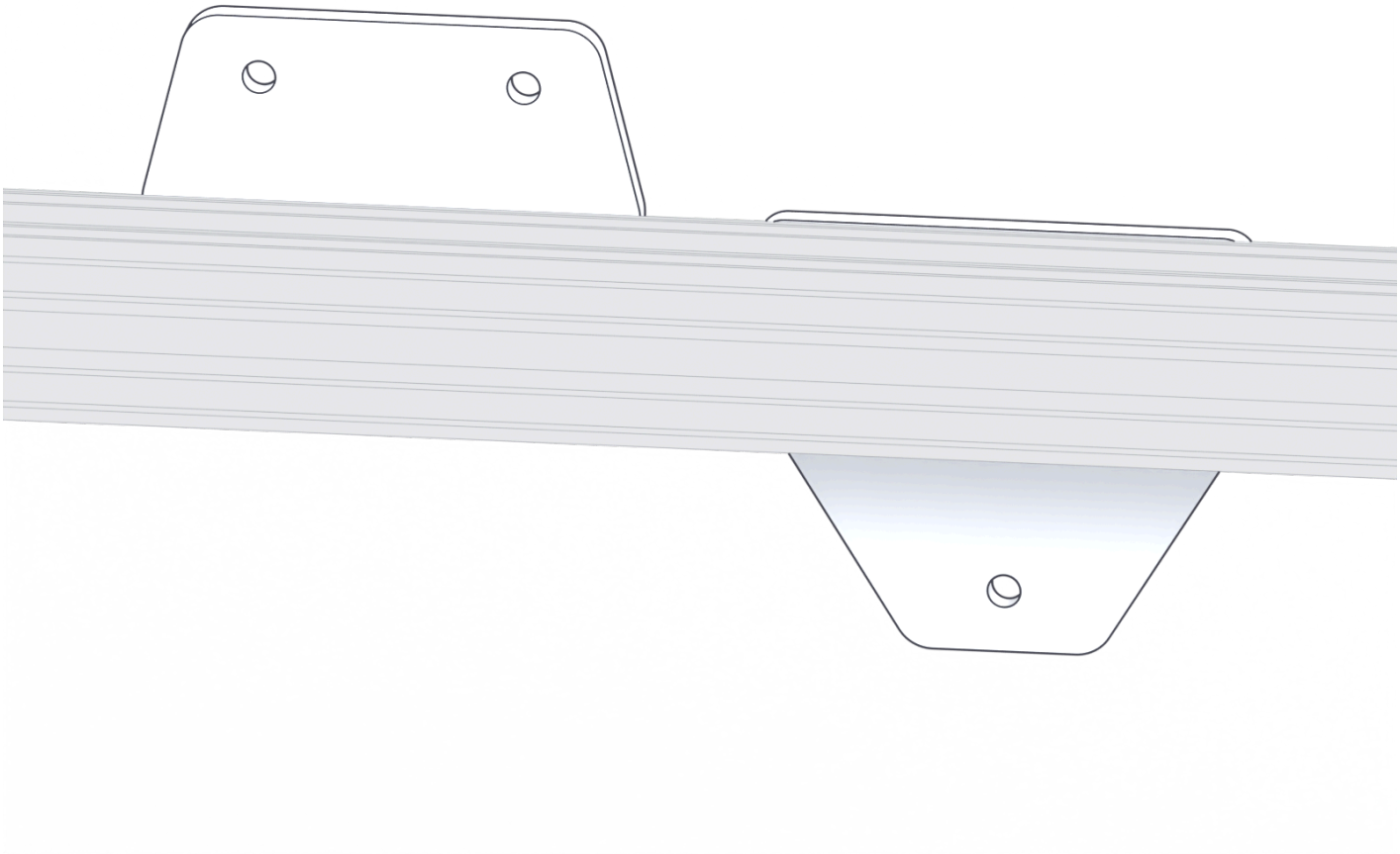
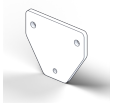
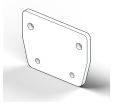


Figure 19: Joint 12: camera_plates, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
15		CAMERA MOUNT PLATE `CAMERA FRAME:1/CAMERA MOUNT PLATE:1`
16		CAMERA MOUNT PLATE - TELEDYNE KINETIX `CAMERA FRAME:1/CAMERA MOUNT PLATE - TELEDYNE KINETIX:1`

Hardware to review at this joint (not an approved BOM)

- S12 Camera, spring and roller plate attachment hardware · candidate, requires review · required unknown (family, not specified); basis: 2 roller plates + 2 spring plates + 2 different camera plates · joint `camera_plates`

Joint 13: spring_roller

context `CAMERA FRAME:1` (faded, 31 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

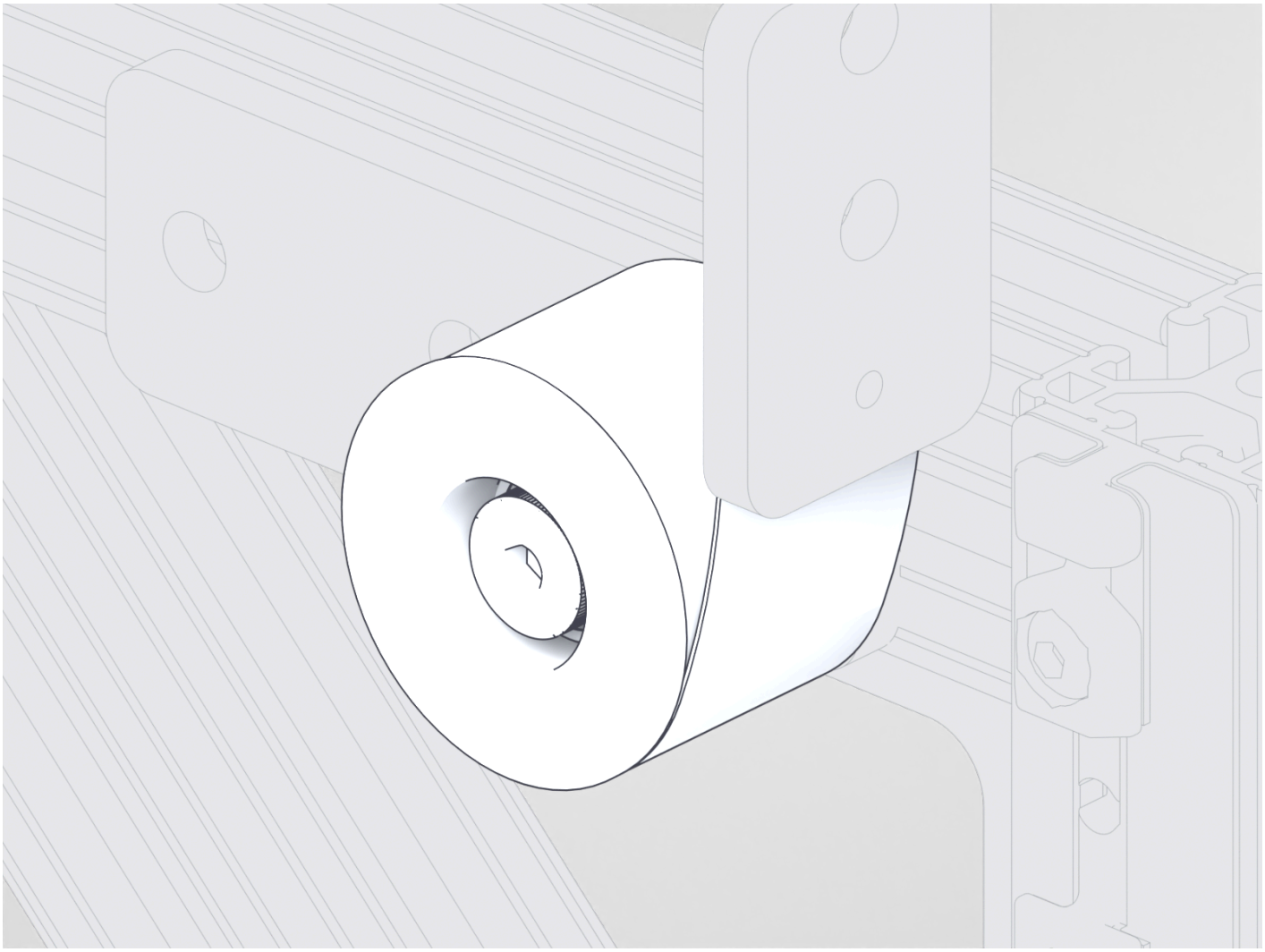
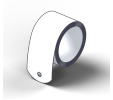


Figure 20: Joint 13: spring_roller, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
12		SPRING ROLLER `CAMERA FRAME:1/SPRING ROLLER:1`
10		MCMASTER 5972K164 Ball Bearing `CAMERA FRAME:1/MCMASTER 5972K164_Ball Bearing:2`
11		MCMASTER 90265A164 18-8 Stainless Steel Shoulder Screw `CAMERA FRAME:1/MCMASTER 90265A164_18-8 Stainless Steel Shoulder Screw:1`
8		MCMASTER 9293K121 Constant-Force Spring `CAMERA FRAME:1/MCMASTER 9293K121_Constant-Force Spring:1`
10		MCMASTER 5972K164 Ball Bearing `CAMERA FRAME:1/MCMASTER 5972K164_Ball Bearing:1`

Hardware to review at this joint (not an approved BOM)

- S13 Spring end and roller/bearing retention hardware · candidate, requires review · required unknown (family, not specified); basis: 2 springs + 2 rollers + 4 bearings + 2 shoulder screws modeled · joint `spring_roller`

Joint 14: camera_attachment

context `CAMERA FRAME:1` (faded, 34 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

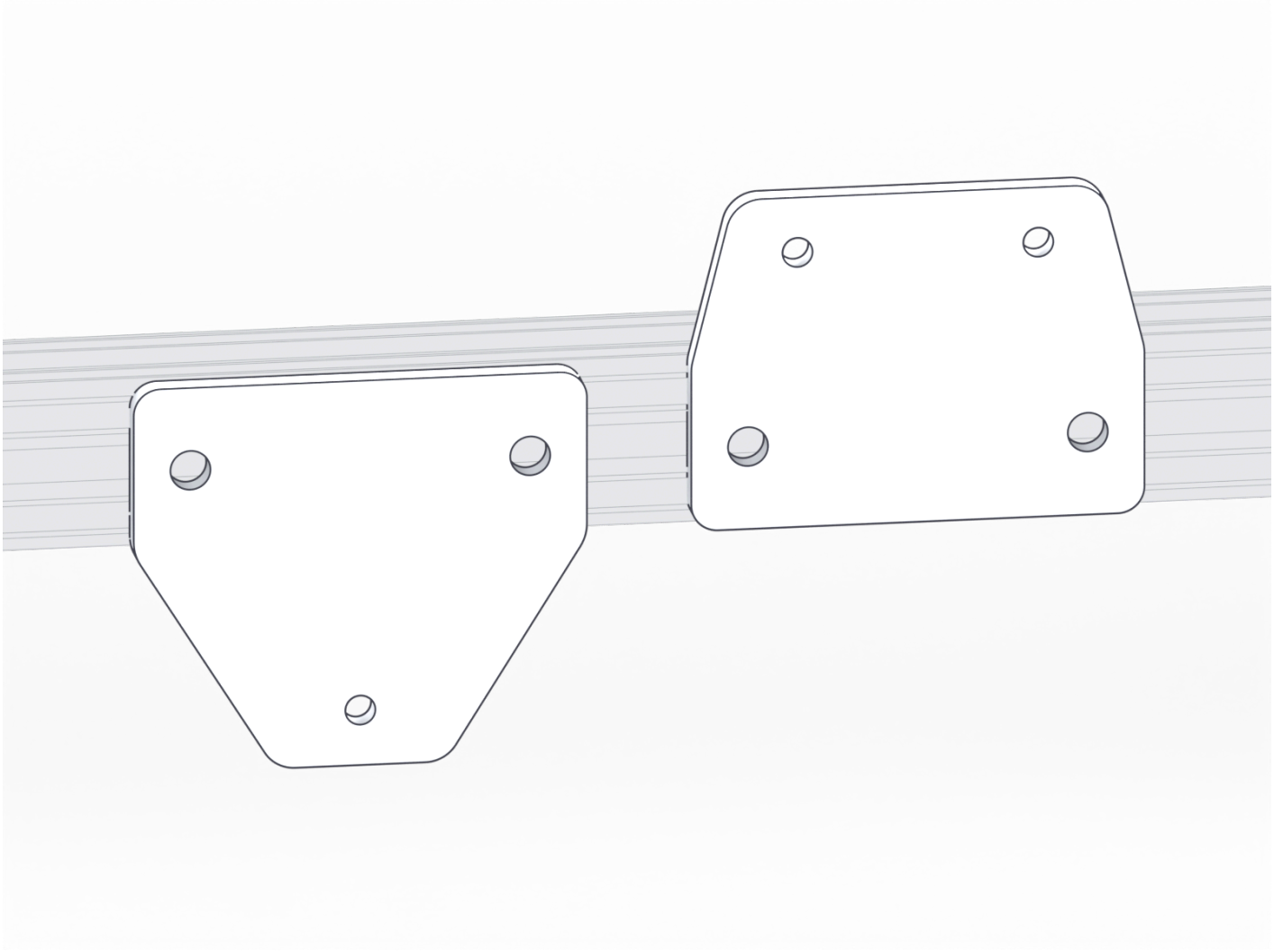
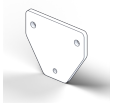
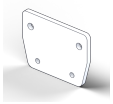


Figure 21: Joint 14: camera_attachment, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
15		CAMERA MOUNT PLATE `CAMERA FRAME:1/CAMERA MOUNT PLATE:1`
16		CAMERA MOUNT PLATE - TELEDYNE KINETIX `CAMERA FRAME:1/CAMERA MOUNT PLATE - TELEDYNE KINETIX:1`

Hardware to review at this joint (not an approved BOM)

- S14 Camera-to-mount attachment · candidate, requires review · required unknown (family, not specified); basis: Camera plate variants present; actual camera selection unconfirmed · joint `camera_attachment`

Joint 15: top_panel

context `root` (faded, 154 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

Figure 22: Joint 15: top_panel, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
34		ENCLOSURE TOP PANEL `ENCLOSURE TOP PANEL:1`

Hardware to review at this joint (not an approved BOM)

- S15 Top-panel retention and edge/support interface · candidate, requires review · required unknown (family, not specified); basis: 1 enclosure top panel · joint `top_panel`

Joint 16: panel_retention

context `FREE STANDING FRAME:1` (faded, 47 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

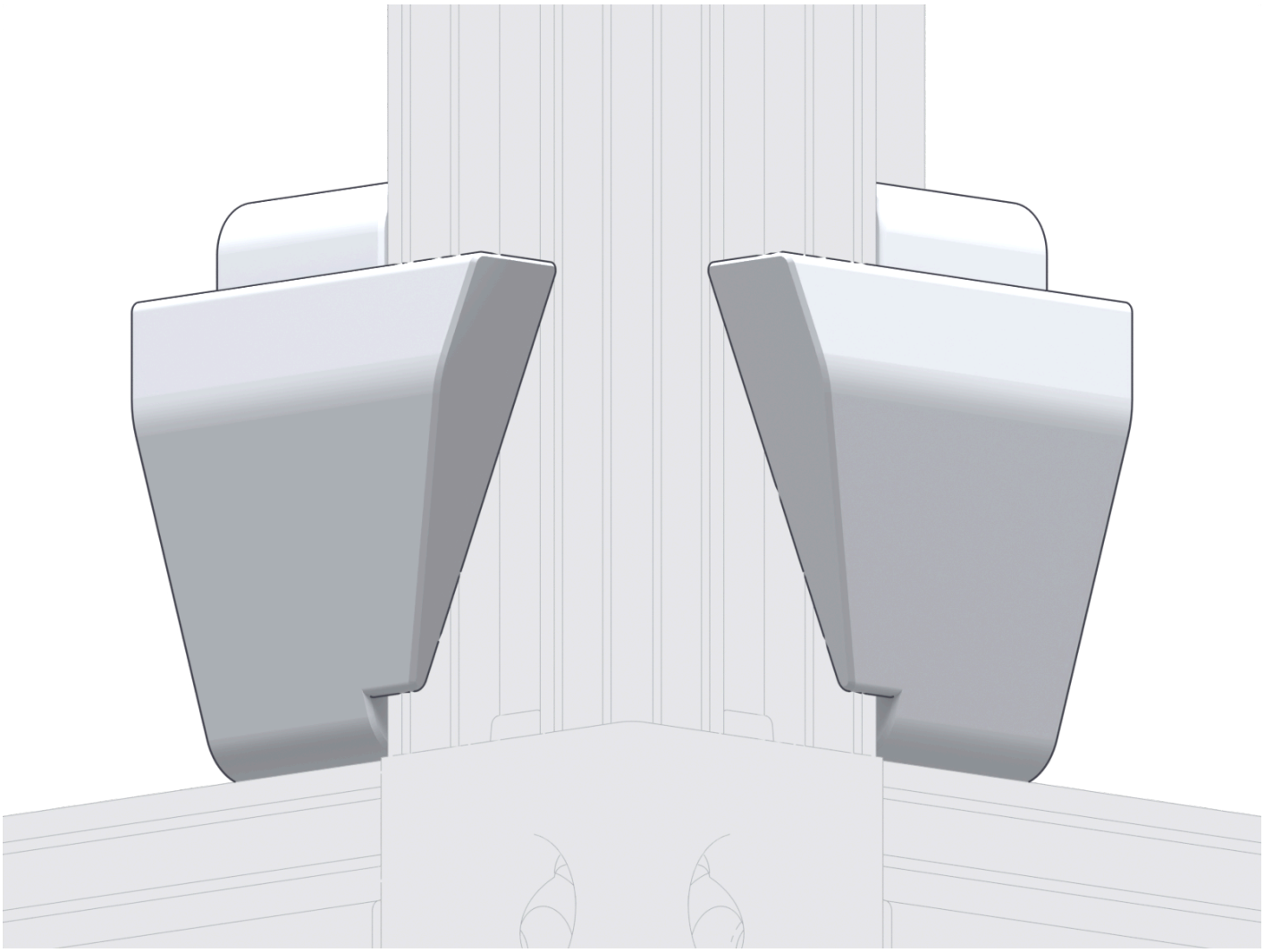


Figure 23: Joint 16: panel_retention, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
30		PANEL LOWER SUPPORT LEFT `FREE STANDING FRAME:1/PANEL LOWER SUPPORT LEFT:1`
31		PANEL LOWER SUPPORT RIGHT `FREE STANDING FRAME:1/PANEL LOWER SUPPORT RIGHT:2`

Hardware to review at this joint (not an approved BOM)

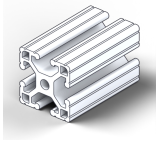
- S16 Removable-panel retention interface · candidate, requires review · required unknown (family, not specified); basis: 4 panels, 8 lower supports, 8 toggle clamps · joint `panel_retention`

Step 1

[bottom-frame]

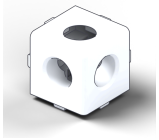
PARTS NEEDED

28 ×2



23 ×3

24 ×4



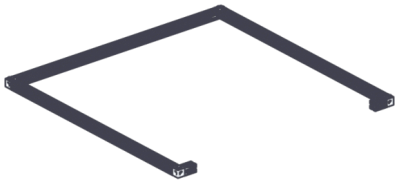


Figure 24: Place as shown.

Lay out the bottom frame: lower rails and corner connectors. Order among rails is free; square the frame before tightening. Hardware: see joints main_frame_joints.

Step 2

[casters] after bottom-frame

PARTS NEEDED

27 ×4

26 ×4

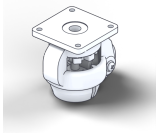


Figure 25: Place as shown.

Bolt each caster to its mounting plate and attach the four plates under the bottom frame corners (joints caster_to_plate, caster_plate_to_frame). Hardware unverified.

Step 3

[columns] after casters

PARTS NEEDED

25 ×4

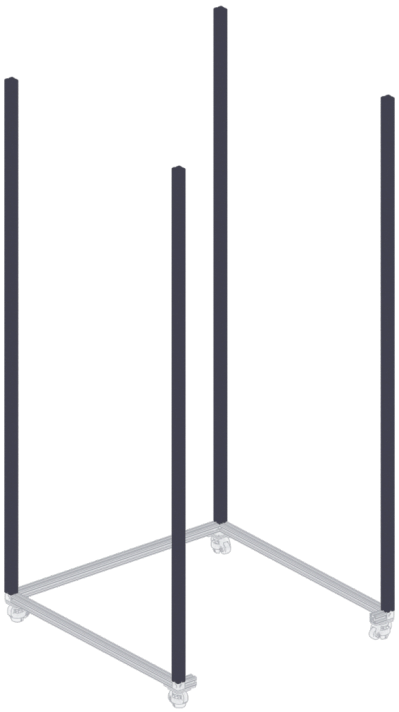


Figure 26: Place as shown.

Stand the four 96 in columns on the bottom frame corners.

Step 4

[middle-rails] after columns

PARTS NEEDED

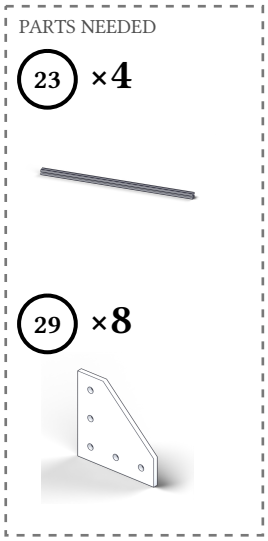


Figure 27: Place as shown.

Add the mid-height rails and extrusions between the columns.

Step 5

[top-frame] after middle-rails

PARTS NEEDED

23 ×4

24 ×4

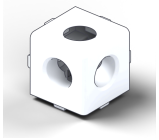


Figure 28: Place as shown.

Close the top frame: upper rails and corner connectors.

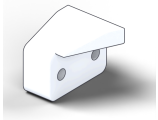
Step 6

[lower-supports] after top-frame

PARTS NEEDED

30 ×4

31 ×4



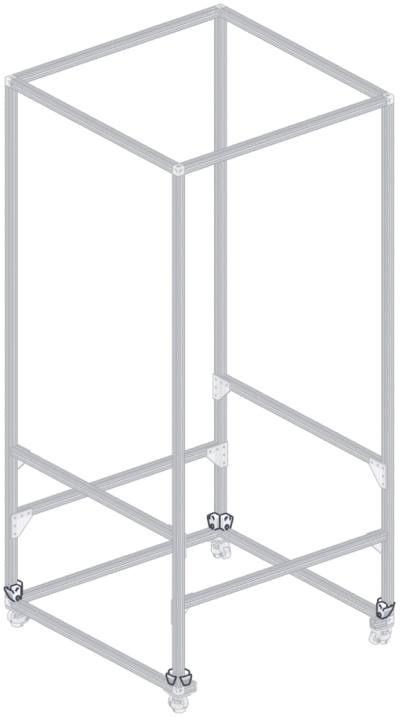


Figure 29: Place as shown.

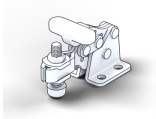
Fit the eight panel lower supports (left and right) to the frame (joint lower_support_to_frame).

Step 7

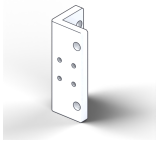
[toggle-clamps] after lower-supports

PARTS NEEDED

35 ×8



36 ×4



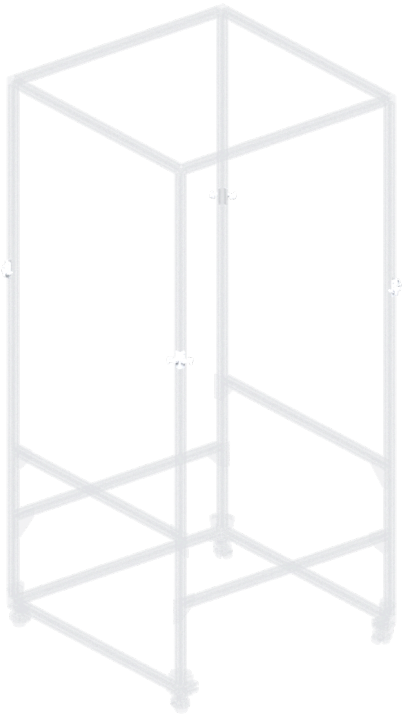


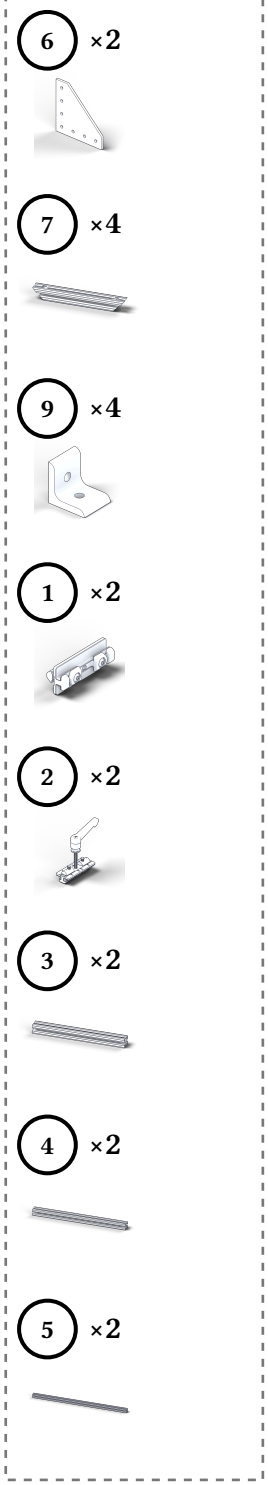
Figure 30: Place as shown.

Assemble the four toggle-clamp mounts (two clamps each) and attach them to the columns (joints clamp_to_mount, clamp_mount_to_frame).

Step 8

[camera-frame-structure] after toggle-clamps

PARTS NEEDED



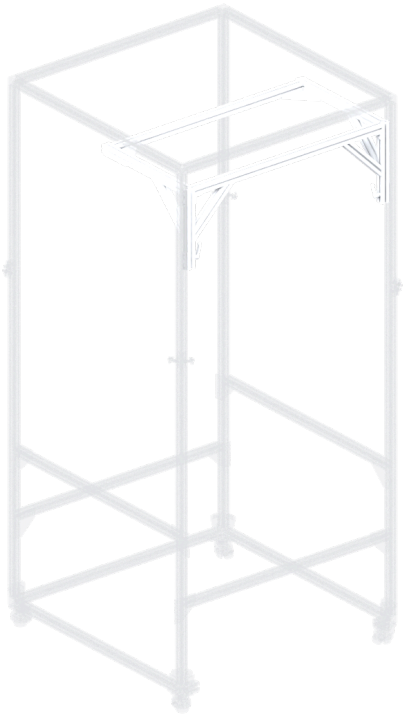


Figure 31: Place as shown.

Assemble the camera frame extrusions and corner brackets (joint camera_frame_joints).

Step 9

[camera-frame-mechanism] after camera-frame-structure

PARTS NEEDED

10 ×4

11 ×2

8 ×2

13 ×2

14 ×2

12 ×2

Figure 32: Place as shown.

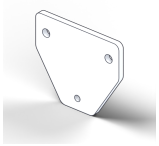
Fit the roller and spring mechanism: roller plates, spring plates, rollers, bearings, shoulder screws, springs (joint spring_roller).

Step 10

[camera-plates] after camera-frame-mechanism

PARTS NEEDED

15 ×1



16 ×1

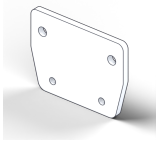


Figure 33: Place as shown.

Fit the camera mount plates (joints camera_plates, camera_attachment). The camera itself is not in CAD.

Step 11

[panel-1] after camera-plates

PARTS NEEDED

32 ×1

33 ×2

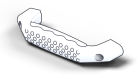


Figure 34: Place as shown.

Fit two pull handles to panel 1 (joint handle_to_panel) and hang the panel on its lower supports and toggle clamps (joint panel_retention).

Step 12

[panel-2] after panel-1

PARTS NEEDED

32 ×1

33 ×2

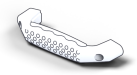


Figure 35: Place as shown.

Fit two pull handles to panel 2 (joint handle_to_panel) and hang the panel on its lower supports and toggle clamps (joint panel_retention).

Step 13

[panel-3] after panel-2

PARTS NEEDED

32 ×1

33 ×2

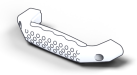


Figure 36: Place as shown.

Fit two pull handles to panel 3 (joint handle_to_panel) and hang the panel on its lower supports and toggle clamps (joint panel_retention).

Step 14

[panel-4] after panel-3

PARTS NEEDED

32 ×1

33 ×2

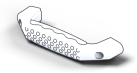


Figure 37: Place as shown.

Fit two pull handles to panel 4 (joint handle_to_panel) and hang the panel on its lower supports and toggle clamps (joint panel_retention).

Step 15

[top-panel] after panel-4

PARTS NEEDED

34 ×1

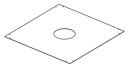


Figure 38: Place as shown.

Place the top panel (joint top_panel: nothing in CAD retains it).

Step 16

[conditional-tank-support-frame-1] after top-panel **CONDITIONAL: install in phase 1 only if it cannot be installed after the frame and panels are assembled; otherwise defer with the tanks**

PARTS NEEDED

20 ×4

19 ×4

18 ×4

22 ×4

21 ×1

17 ×4

Figure 39: Place as shown.

CONDITIONAL (build-scope.json): TANK SUPPORT FRAME:1, the tank support frame. Install now only if it cannot be installed after the panels; otherwise defer with the tanks. Joints support_frame_corners, support_brackets, vibration_mounts.

Step 17

[conditional-tank-support-frame-2] after conditional-tank-support-frame-1 **CONDITIONAL: install in phase 1 only if it cannot be installed after the frame and panels are assembled; otherwise defer with the tanks**

PARTS NEEDED

20 ×4

19 ×4

18 ×4

22 ×4

21 ×1

17 ×4

Figure 40: Place as shown.

CONDITIONAL (build-scope.json): TANK SUPPORT FRAME:2, the tank support frame. Install now only if it cannot be installed after the panels; otherwise defer with the tanks. Joints support_frame_corners, support_brackets, vibration_mounts.

Deferred steps (not in this build)

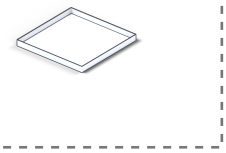
These steps place items outside phase-1; they are shown seated on the finished build for completeness and are not part of the build sequence above.

Step 18

[deferred-tanks] after conditional-tank-support-frame-2 **DEFERRED: not in this build**

PARTS NEEDED

37 ×3



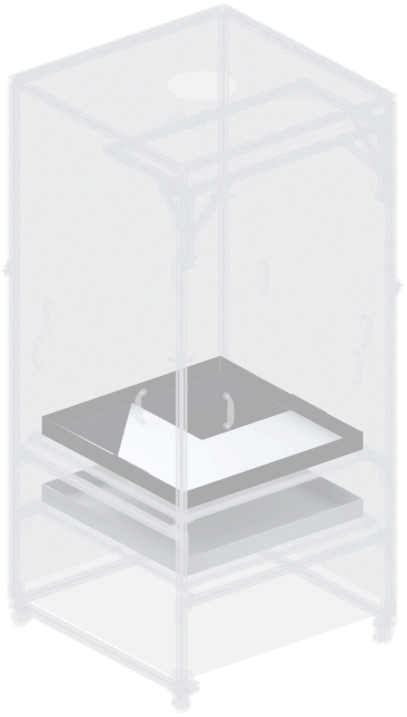


Figure 41: Place as shown.

DEFERRED (build-scope.json): the tanks are not part of phase 1 and are listed here only because every CAD occurrence must appear once. They are to be replaced by monitors and bowls.

Build record

generated	2026-10-23T15:40:18Z
scene	free-standing-enclosure-991056: Free Standing Enclosure (Vault file-version 991056) native snapshot (37 parts, 155 occurrences, 17 assemblies)
source kind	autodesk-inventor-vault-snapshot
source root_version_id	991056
source exporter	windows/Export-PartAssets.ps1 on Inventor 2025.3
source qualification	geometry/hierarchy snapshot; not a reviewed assembly procedure or stock inventory
source occurrence_id_scope	snapshot-scoped occurrence paths (Inventor occurrence names); stable only within this snapshot
scene.json	`ac203e5656abdeeadbadf5f047e31f244e01f8938001c5747a44808ffc4ed070`
instructions.json	`9de40bb2e8925e5f456645efd9c465bfc3576103fd1ac0d4f3bf45e6d5c6d3f5`
inventory.json	`0eed5897bb238c1334e92e19927db8573480f3c7d4778099fa02b09e605298de`
supplement.json	`c9950ec3f72db98f8d3563adaec0fb48ba50fce689fbb905dbe050ce0e4081ba`
build-scope.json	`b835cb40d41f9b05eadb57ca58dfdaf7ae0a05deddb1c951bcc009325f2d4911`
assets	37 STEP files; hashes in manual.json
FreeCAD	1.1.3; deflection 0.05 mm / 10.0°, budget 50000 triangles per part; meshes.json `91c004502b7a089aac81b6d07810e5eae461b3174bfaaca41c468ba3a5a6f414`
Blender	4.5.14 LTS, engine CYCLES, denoiser OPENIMAGEDENOISE, samples 64, long side 1600 px; line art on up to 600000 visible triangles; settings `94534a5410b48956efa4f48c2626ab3b0f08ae3cb5c1c6c6495fc0dc789e620b`
reused views	68 of 86 images copied from an earlier run with the same inputs, meshes and engine (build/enclosure-snapshot-991056/renders10/render-receipt.json); per-view record in manual.json
render warning	step-07-toggle-clamps: 909744 visible triangles exceed lines.max_triangles (600000); rendered without line art
render warning	step-08-camera-frame-structure: 1180312 visible triangles exceed lines.max_triangles (600000); rendered without line art
render warning	step-09-camera-frame-mechanism: 1379408 visible triangles exceed lines.max_triangles (600000); rendered without line art
render warning	step-10-camera-plates: 1382084 visible triangles exceed lines.max_triangles (600000); rendered without line art
render warning	step-11-panel-1: 1473284 visible triangles exceed lines.max_triangles (600000); rendered without line art
render warning	step-12-panel-2: 1564484 visible triangles exceed lines.max_triangles (600000); rendered without line art
render warning	step-13-panel-3: 1655684 visible triangles exceed lines.max_triangles (600000); rendered without line art
render warning	step-14-panel-4: 1746884 visible triangles exceed lines.max_triangles (600000); rendered without line art
render warning	step-15-top-panel: 1749932 visible triangles exceed lines.max_triangles (600000); rendered without line art
render warning	step-16-conditional-tank-support-frame-1: 2092304 visible triangles exceed lines.max_triangles (600000); rendered without line art
render warning	step-17-conditional-tank-support-frame-2: 2434676 visible triangles exceed lines.max_triangles (600000); rendered without line art
render warning	step-18-deferred-tanks: 2436488 visible triangles exceed lines.max_triangles (600000); rendered without line art
images	86 renders; hashes in manual.json
supplement.json	20 row(s): 16 existing design, 4 new design; never added to CAD quantities
generator	python 3.12.12; platform Linux-6.12.113+deb13-amd64-x86_64-with-glibc2.41; package mechanical-docs

Review state draft; physical assembly not validated.